

Zombie Firms in Emerging Markets: Survival and Funding Mechanisms in the Indian Context

ABSTRACT

The study analyses the identification, financial traits, and life cycle of zombie firms in India's non-financial sector. The study makes several contributions to understanding zombie firms in India's non-financial sector. First, it identifies zombie firms and tracks their financial characteristics and life-cycle dynamics, showing that although they constitute a small fraction of the firm population, they hold a disproportionately large share of total debt and assets, with potential implications for financial stability and economic efficiency. Second, it shows that signs of financial deterioration are evident well before zombification, and that zombie firms exhibit increasing dependence on debt financing, along with limited improvement in core performance indicators. Third, it highlights the association between the funding structure and zombie-firm outcomes. Firms primarily financed through equity are associated with a lower likelihood of zombie status, a higher likelihood of recovery, and a lower likelihood of relapse. In contrast, firms primarily financed through bank borrowing are associated with a higher likelihood of zombie status, longer periods of distress, and a greater likelihood of relapse. Finally, the study shows that a higher number of banking relationships is positively associated with zombie status, while promoter pledging is also positively associated with zombie status. These patterns are consistent with the possibility that financially distressed firms rely more heavily on external sources of finance, although the underlying mechanisms cannot be established from the present analysis

Keywords: Zombie Firms, Zombie Lifecycle, Corporate Finance, Banks, India

JEL Codes: C41, D25, E51, G32, G33, L25

1. INTRODUCTION

Corporate finance literature has widely examined firm performance in relation to capital structure and survival during periods of financial distress (Modigliani & Miller, 1958; Myers, 1984). A core challenge in this field, however, is accurately distinguishing between healthy and distressed firms—an essential task for informed investment, effective policy design, and the maintenance of macroeconomic stability (Altman, 1968). Identifying distressed firms is thus critical not only for economic policymaking but also for practical decisions involving credit, investment, and bankruptcy. Recently, research attention has increasingly turned to a distinct subset of distressed firms—so-called “zombie firms”—which pose significant risks to financial stability and economic productivity, particularly in emerging markets with evolving regulatory frameworks. Zombie firms are generally defined as businesses that fail to generate sufficient profits to cover their debt servicing costs while continuing to operate due to ongoing external support (Caballero, Hoshi, & Kashyap, 2008). They are typically characterised by persistently low-interest coverage ratios (ICRs), weak balance sheets, and frequent debt rollovers (McGowan et al., 2018; Storz et al., 2017; Fukuda & Nakamura, 2011). However, definitions and identification criteria vary across studies, reflecting differences in how financial distress is measured (Banerjee & Hofmann, 2022; Schivardi et al., 2022).

Studying zombie firms is essential due to their broader implications for economic efficiency, financial stability, and resource allocation. By continuing to operate without sustainable profitability, these firms often distort market signals and impede the reallocation of capital and labour to more productive uses. Their prolonged survival is often supported by accommodative credit conditions, government interventions, or banking forbearance, which can pose systemic risks if not addressed (McGowan et al., 2018). However, zombie status is not necessarily permanent—some firms recover through restructuring or improved market conditions, while others relapse after temporary improvement, raising concerns about the durability of their recovery. Understanding these divergent paths is critical for evaluating their long-term economic impact. Recent studies (Banerjee and Hofmann (2022), Tuuli (2024)) highlight that zombie firms exhibit distinct life-cycle patterns, including phases of recovery, persistence, or eventual exit, and underscore the evolving role of both bank and non-bank financial institutions in sustaining zombies at different stages of distress. These studies further show that sources of financing influence zombie-firm dynamics across different stages of the life cycle. For example, Tuuli (2024) finds that equity funding becomes more prominent in later stages as banks retreat, highlighting how the source of finance shapes firms' recovery and persistence.

While global research has increasingly adopted a life-cycle perspective on zombie firms, important questions remain about how financing structures shape zombie dynamics across different institutional settings. Banerjee and Hofmann (2022) document the anatomy and life cycle of zombie firms across advanced economies, showing that firms differ in their persistence, recovery, and relapse patterns. Extending this perspective, Tuuli (2024) demonstrates that sources of financing play an important role in these transitions, with bank and non-bank funding contributing differently across the zombie life cycle. However, these studies are conducted in advanced economies with institutional environments that differ substantially from those of bank-dominated emerging markets. In India, existing research has primarily focused on identifying zombie firms and examining their macroeconomic implications (Kulkarni et al., 2021; Pattanaik et al., 2022; Rai, 2025), leaving relatively little evidence on how sources of financing influence the persistence, recovery, relapse, and survival of zombie firms over time. This study addresses this gap by adopting a life cycle perspective to examine how alternative funding sources shape zombie-firm dynamics in the Indian institutional context.

The Indian institutional setting provides a particularly relevant context for examining zombie-firm dynamics due to the dominant role of banks in corporate financing, the prevalence of consortium-based lending structures, and the concentration of promoter ownership in listed firms. Unlike in market-based financial systems, where firms rely more heavily on capital markets, Indian firms remain significantly dependent on bank credit, which may increase the likelihood of prolonged refinancing and delayed restructuring for financially weak firms. In addition, consortium lending arrangements may weaken lender coordination and monitoring incentives, potentially encouraging continued support to distressed firms. Similarly, the dominance of promoter-led governance structures in India makes promoter share pledging an important indicator of financial fragility and distress. These institutional features make India a particularly relevant setting for examining how financing structure, lender relationships, and governance characteristics shape zombie-firm persistence, recovery, and relapse dynamics. Accordingly, the institutional characteristics of the Indian financial system suggest that the relationship between financing sources and zombie-firm outcomes may differ from that documented in advanced economies, providing the basis for the empirical analysis undertaken in this study.

Beyond examining the role of financing sources across the zombie firm life cycle, the study also extends the literature by incorporating two firm-specific characteristics that are

particularly relevant to the Indian institutional setting: the number of banking relationships and promoter share pledging. Prior studies have largely examined the role of banks in sustaining zombie firms through regulatory forbearance and lending relationships. Building on this literature, the present study investigates whether the extent of firms' banking relationships is associated with zombie status in an institutional environment where consortium lending is common, and lending decisions may be influenced by coordination failures among multiple banks. In addition, the study examines promoter share pledging as a governance-related characteristic that has received limited attention in the zombie firm literature. Given the prevalence of concentrated promoter ownership in India, promoter pledging may signal financial fragility and increased dependence on external finance, making it a potentially important correlate of zombie status. By incorporating these institutional features into a life-cycle framework, the study provides new evidence on how financing arrangements and governance characteristics are associated with zombie-firm persistence, recovery, and relapse in an emerging-market economy.

The paper finds that bank-financed firms are more likely to become zombies, remain distressed longer, and face a higher risk of relapse after recovery. In contrast, equity-funded zombies recover more quickly and relapse less often. However, even recovered firms tend to show weak financial health, with low interest coverage, high leverage, and modest returns—implying that recovery does not mean full financial revival, a result similar to that of Banerjee and Hofmann (2022). Further, the study shows that a greater number of banking ties is positively associated with zombie status, likely due to easier debt rollovers stemming from herd behaviour and weaker lender discipline. Promoter pledging is also associated with higher zombification risk, suggesting it may signal distress rather than stability. These findings are especially pertinent to India, where corporate capital structures are heavily tilted towards debt—primarily sourced from the banking sector—and undercapitalised banks remain significantly exposed to chronically weak firms, thereby heightening risks to overall financial stability (Acharya et al., 2019). The remainder of the paper is organised as follows: Section 2 reviews the relevant literature on zombie firms in India and internationally. Sections 3 and 4 outline the data and methodology used in the analysis. Section 5 presents the empirical results. Finally, Section 6 concludes with the findings and policy implications.

2. LITERATURE REVIEW

The issue of zombie firms gained prominence in the context of Japan's post-bubble economy in the 1990s. Caballero, Hoshi, and Kashyap (2008) introduced the term “zombie firms” to describe non-viable companies sustained by banks reluctant to recognise loan losses. By using interest rate gaps to identify these firms, they demonstrated how bank forbearance distorted market signals, depressed productivity, and obstructed economic restructuring. Fukuda and Nakamura (2011) later examined the role of restructuring in addressing zombification, showing that measures such as asset sales and labour reductions could aid recovery, but only under favourable macroeconomic conditions. These early studies firmly established the link between weak financial institutions, regulatory forbearance, and the survival of unproductive firms.

The debate soon extended to European economies, where similar patterns emerged. Storz et al. (2017), in a study of five European countries, found that undercapitalised banks delayed deleveraging, thereby enabling firms with sustained negative returns and poor debt-servicing capacity to persist. Acharya et al. (2019) reinforced this view in the context of GIPS nations, arguing that politically influenced and weakly capitalised banks continued lending to distressed borrowers, thereby crowding out healthy firms and dampening job creation. McGowan, Andrews, and Millot (2018) broadened the scope to include OECD countries and defined zombies as firms at least 10 years old and unable to cover interest costs for 3 consecutive years. Their work highlighted the broader economic costs of zombification, including suppressed new-firm entry, inflated wage-to-productivity ratios, and weakened market dynamism.

However, recent studies have shifted focus from macroeconomic distortions to firm-level dynamics. An essential contribution in this direction is by Banerjee and Hofmann (2022), who analysed the internal characteristics and life cycles of zombie firms. By incorporating both accounting- and market-based indicators, they offered a more refined classification that considers not only historical performance but also future viability. Importantly, their longitudinal approach showed that exiting zombie status does not necessarily imply full recovery, as many such firms continued to underperform.

Building on earlier work, Tuuli (2024) advances the literature by refining zombie firm identification and lifecycle analysis. Unlike studies based solely on accounting data, Tuuli incorporates survey data to distinguish between temporarily distressed and truly unviable firms. Using survival analysis, the study adopts a dynamic approach to track transitions into and out of zombie status, revealing that recovery—not bankruptcy—is the more common exit path. Tuuli further explores the capital structure decisions of these firms, revealing that while banks remain essential funders, equity financing is associated with prolonged zombification.

Meanwhile, the discourse on zombie firms has gained renewed relevance in India, where financial sector inefficiencies and regulatory leniency have exacerbated the problem. Chopra, Subramanian, and Tantri (2020) examined the Reserve Bank of India's Asset Quality Review (AQR) and found that, while it improved transparency, undercapitalised banks continued lending to distressed firms—indicating persistent gaps in credit discipline. This finding suggested that India's regulatory measures, although helpful in some respects, had not yet fully addressed the underlying issue of zombie-firm financing. They also pointed out that regulatory forbearance post-global financial crisis had encouraged Indian banks to adopt an “extend and pretend” approach, rolling over bad loans rather than recognising them. These practices allowed distressed firms to continue operating, further delaying their necessary exit from the market. The authors argued that such regulatory accommodation created a misalignment between firms' viability and access to credit, worsening the issue of zombie firms. Pattanaik, Muduli, and Jose (2022) estimated that zombie firms in India account for nearly ten percent of total non-financial corporate debt, often exhibiting high leverage and persistently negative returns. These firms primarily borrow to survive rather than invest in growth, and the similarity of credit terms extended to both zombie and healthy firms reflects major flaws in credit risk assessment. Contrary to trends in advanced economies, Pattanaik et al. (2022) found that India's accommodative monetary policy post-2014 did not significantly promote zombie lending. Rather, they argued that low interest rates benefited productive firms. Nevertheless, the persistence of zombie firms during this period pointed to deeper institutional and structural inefficiencies in credit allocation and corporate governance. Capital structure has also emerged as a critical factor in zombie firms' survival and fragility. Asha Rai (2025) found that Indian zombie firms exhibit extraordinarily high debt-to-equity ratios—up to twelve times those of healthy firms. These firms rarely issue new equity, likely due to low investor confidence and weak retained earnings, relying instead on additional borrowing. Rai and Sharma (2023) extended the analysis by showing that zombies were concentrated in cyclical, low-entry-barrier

sectors such as construction and trade, where poor profitability, weak asset returns, and excessive leverage were common. These findings underscore how poor financial planning and sectoral vulnerabilities compound zombification.

Despite these valuable insights, much of the existing literature on zombie firms in India remains focused on identifying zombie existence and understanding their macro-level determinants, with limited exploration into how these firms evolve or exit their distressed state. Moreover, there is little empirical work examining the role of funding composition—especially the differential effects of bank versus equity financing—on these transitions. Existing research also fails to acknowledge potentially informative firm-level variables, such as the number of banking relationships or promoter share pledging, which could reflect deeper patterns of credit dependence or governance-induced financial distress

The Indian financial system is characterised by a dominant role of banks in corporate financing, with state-owned banks accounting for a large share of credit to the non-financial sector. Limited access to market-based financing and persistent reliance on bank credit relationships make firms particularly reliant on bank lending, even during periods of financial distress. Historical tendencies toward loan forbearance and delayed recognition of non-performing assets have contributed to the persistence of financially weak firms. Moreover, promoter ownership structures and the widespread practice of share pledging create incentives for firms to survive even during prolonged distress. The introduction of the Insolvency and Bankruptcy Code (2016) and subsequent regulatory reforms have begun to alter the resolution landscape, but the institutional environment continues to shape the financing behaviour and survival dynamics of distressed firms.

To address these gaps, this study takes a lifecycle approach to explore the determinants of funding for zombie firm survival, recovery, and relapse in India, offering new evidence on how financing patterns influence firm outcomes over time.

3. DATA AND SAMPLE

The study primarily draws data from CMIE Prowess, a comprehensive financial database that includes both publicly traded and unlisted firms. Prowess has been widely used in previous research¹ to assess the financial health of companies. Our analysis focuses on the sample period of 2002 to 2023 and is restricted to non-financial companies. To identify zombie firms, the study tracks various financial characteristics over time using firms' financial statements and reports. After excluding firms with missing values, the final sample comprises 30,895 unique firms, yielding 288,687 firm-year observations. To avoid misclassifying early-stage start-ups as zombie firms, the study applies a threshold requiring firms to have been operational for at least five years. This criterion further narrows the sample size to 269,320 firms. Following the existing literature, the study employs financial indicators such as the Interest Coverage Ratio (ICR), Debt-to-Assets ratio, Debt Growth, Net Profit Margin, and Sales Growth to identify zombie firms.

In addition to these core identifiers, data related to borrowing behaviour (e.g., borrowing from banks and financial institutions, inter-corporate loans, subsidised credit), promoter pledging, and the number of banking relationships were used to analyse the factors influencing zombification, recovery, continuation, and relapse. A detailed list of variables is provided in Appendix A. Finally, all continuous variables are winsorized at the top and bottom 1% of their firm-year distributions to mitigate the influence of outliers.

¹ Asha Rai (2025), Rai and Sharma (2023), and Kulkarni et al. (2021)

4. METHODOLOGY:

4.1 Identification of Zombie Firms

Our methodology for identifying zombie firms draws on a broad body of existing literature. Early studies identified zombie firms using interest rate gaps (Caballero et al., 2008) and later incorporated profitability and leverage measures (Fukuda & Nakamura, 2011). Subsequent research added indicators such as debt growth, credit ratings, and debt-servicing capacity. Across the literature, zombie firms are consistently marked by low-interest coverage, persistent losses, weak balance sheets, and reliance on continuous debt refinancing (McGowan et al., 2018; Banerjee & Hofmann, 2022). Consistent with the existing studies, we define zombie firms using three distinct criteria that capture varying levels of financial distress.

4.1.1 Weak Definition

Under a weak definition, zombie firms are identified using less stringent thresholds to capture a broader set of financially weak but still operating firms. A firm is classified as a zombie if it meets all of the following criteria: (1) an Interest Coverage Ratio (ICR) below 1, indicating inability to cover interest expenses from operating income; (2) leverage above the industry median, reflecting high debt dependence; (3) positive debt growth, suggesting continued borrowing despite weak performance; and (4) a net profit margin below the industry median, signalling sustained underperformance.

4.1.2 Semi-Strong Definition

These firms often rely on ongoing financial support from banks, creditors, or forbearance policies, which can create market inefficiencies by misallocating capital and labour away from more productive enterprises. Under this definition, a firm is classified as a zombie if it meets all of the following criteria: (1) Interest Coverage Ratio (ICR) below 1; (2) leverage above the industry median; (3) positive debt growth; (4) net profit margin below the industry median; and (5) negative sales growth.

4.1.3 Strong Definition

To account for persistency, a firm is classified as a zombie under this definition if it meets all of the following conditions for two consecutive years: (1) an Interest Coverage Ratio (ICR) below 1; (2) leverage above the industry median; (3) positive debt growth; (4) a net profit margin below the industry median; and (5) negative sales growth.ⁱ

For all subsequent analyses, we use the semi-strong definition, which balances the identification of at-risk firms with the need for non-overly-strict duration criteria. By adding negative sales growth, it captures operational decline and enables earlier detection than the more restrictive two-year standard, offering a more accurate and timely view of distress. However, we have conducted robustness checks² using the other definitions to ensure the consistency and reliability of our findings.

4.2 Classification of Primary Sources of Funding

To identify the primary source of funds, the study applies a threshold where a funding source must constitute at least 50% of a firm's total assets in a given year. If no single source meets this threshold, the firm is classified as "Other-funded," indicating a mixed financing structure. Based on this, firms are categorised into three mutually exclusive groups: bank-funded, equity-funded, and other-funded. Note, that the use of a 50% threshold aligns with established corporate finance literature, which defines dominant ownership, funding, or control based on the majority (La Porta et al., 1999; Faccio & Lang, 2002; Almeida & Wolfenzon, 2006). This approach offers a clear and tractable measure of primary funding dependence, reducing classification ambiguity. It also reflects institutional norms in financial reporting and empirical research, where a simple majority is commonly used to signify influence or control (Lemmon & Roberts, 2010).

4.3 Performance of Firms Before and After Zombification

Next, to explore the dynamics of zombification, the study employs a local projection technique, which traces the evolution of a firm's financial performance before and after entering zombie status (Jorda, 2005). This allows for distinguishing between the immediate impact of zombification and its prolonged effects. By incorporating firm and time-fixed effects, the analysis controls for unobserved heterogeneity and common shocks, enhancing the robustness of the estimates. This dynamic approach provides insights into how and when zombie status influences critical performance indicators such as leverage, interest coverage, cash flow, capital expenditure, and asset disposal and certain financing indicators such as borrowings from banks, borrowings from financial institutions, Inter-corporate loans, loans from promoters, directors

² As a robustness check, the semi-strong definition was additionally restricted to firms with ICR < 1 for three consecutive years. The results remain qualitatively similar to those obtained under the strong definition, indicating that the findings are robust to alternative persistence-based specifications.

and shareholders, number of banking relations and amount of promoter pledging. The local projection model is an econometric approach used to estimate the dynamic response of a dependent variable to a shock over multiple time horizons. Instead of estimating a full system of equations as in vector autoregressions (VARs), local projections estimate the impulse response at each horizon h directly using separate regressions of the form:

$$y_{i,t+h} = \alpha_i + \lambda_t + \beta_h \cdot \text{EntryZombie}_{i,t} + \gamma_h \cdot \text{OngoingZombie}_{i,t} + \varepsilon_{i,t+h} \quad (3.1)$$

Where, $y_{i,t+h}$ represents the outcome variable of firm i , at time t , α_i represents firm-specific fixed effects, and λ_t represents time-specific fixed effects. β_h captures the effect of the firm entering zombie status at time t on the outcome variable at time $t+h$, while γ_h indicates the effect of being an ongoing zombie on the outcome variable at time $t+h$, and $\varepsilon_{i,t+h}$ is the error term. The coefficients β_h and γ_h capture the dynamic effects of entering zombie status and remaining a zombie, respectively, on the outcome variable at different horizons h (e.g. from -4 to +4 years).

4.4 Determinants of Zombie Status

To analyse the determinants of zombie status, the study employs a random-effects logistic regression model for firm-level panel data. The model accounts for unobserved firm-specific heterogeneity while allowing estimation using both within- and between-firm variation (Greene, 2004; Allison, 2009). This approach captures persistent, firm-specific factors influencing zombification and offers a robust framework for examining its covariates across heterogeneous firms (Wooldridge, 2010; Hsiao, 2005), as corroborated by prior studies (Banerjee & Hofmann, 2018; McGowan et al., 2018).

To mitigate concerns regarding reverse causality, all explanatory variables are lagged by one period. The baseline model is specified as follows:

$$\text{Logit}(\text{Pr}(Z = 1)) = \alpha_s + u_i + \alpha_t + \beta_1 \text{Primary Bank}_{i,t-1} + \beta_2 \text{Primary Equity}_{i,t-1} + \varepsilon_{it} \quad (3.2)$$

The Zombie (Z) is a binary variable equal to 1 if firm (i) is classified as a zombie firm in year (t), based on the Semi-Restricted definition, and zero otherwise. The primary explanatory variables are $\text{Primary Bank}_{i,t-1}$ and $\text{Primary Equity}_{i,t-1}$, which are dummy variables indicating whether the firm's main source of external finance in year $t-1$ is bank borrowing or equity, respectively. The reference category includes other sources such as Intercompany loans,

nonbank financial institutions, and borrowings from the government. The model specification includes sector fixed effects (α_s) to control for sector-specific influences, time fixed effects (α_t) to control for time-specific influences, and firm-level unobserved heterogeneity, captured through random effects (u_i), which are assumed to be uncorrelated with the explanatory variables. Although fixed-effects logit is a common alternative for binary panel models, it identifies coefficients only from firms that experience changes in zombie status over time. Consequently, firms with no within-firm variation in the dependent variable are excluded from estimation. In our sample, this reduces the estimation sample from 264,988 observations (23,637 firms) to 76,480 observations (6,244 firms), corresponding to a loss of approximately 71.1% of observations and 73.6% of firms. We therefore use the random-effects logit model as the baseline specification to avoid information loss. However, to assess the robustness of our findings to the random-effects assumption, we additionally estimate a correlated random-effects (Mundlak, 1978) specification. Further, we append the baseline specification by adding a set of control variables that includes the number of banking relationships and promoter share pledging, along with other conventional firm-specific variables such as current ratio, log of total assets for size, earnings to debt ratio, asset disposal, return on assets, cash flow, age, and business group affiliation.

$$\begin{aligned} \text{Logit}(\text{Pr}(Z = 1)) = & \alpha_s + \alpha_t + \beta_1 \text{Primary Bank}_{i,t-1} + \beta_2 \text{Primary Equity}_{i,t-1} + \\ & \beta_3 \text{Number of Banks}_{i,t-1} + \beta_4 \text{Promoter pledging}_{i,t-1} + X'_{i,t-1} \gamma + u_i + \varepsilon_{it} \end{aligned} \quad (3.3)$$

The inclusion of firm-level controls follows the established literature on zombie firms, as these variables capture firm-specific financial characteristics and institutional factors that may influence the likelihood and persistence of zombie status. In the specification, $X'_{i,t-1}$ represents additional lagged firm-level controls, and ε_{it} is the error term. The following table lists the control variables included in model 3.3. Finally, to specifically examine the determinants of zombie status among listed firms, the study introduces an additional model that includes firm-level controls available only for publicly traded firms. This enables an investigation into whether the influence of financial structure and firm characteristics differs between listed firms and the broader sample. These additional controls include the Altman Z-Score for financial distress and bankruptcy risk, the Debt-to-equity ratio, reflecting the firm's reliance on debt financing, and Return on Equity to assess efficiency.

Table 1: List of Control Variables

Control Variable	Explanation
Current Ratio	Captures short-term solvency and operational & financial health of a firm.
Log of Total Assets	Controls for firm size, which influences access to finance and resilience to shocks.
Earnings to Debt Ratio	Indicates debt-servicing ability, a central feature of zombie firms.
Asset Disposal	Acts as a proxy for financial restructuring, common among zombie firms.
Return on Assets (ROA)	Reflects how efficiently a firm uses its assets to generate profit.
Cash Flow	Reflects internal fund generation capacity, essential for sustainable operations.
Firm Age	Controls for firm maturity.
Business Group	Dummy to express whether a firm belongs to any business group

4.5 Life Cycle of Zombie Firms

Beyond analysing the determinants of zombie status, the study utilises the Kaplan-Meier estimator to estimate the duration of continuation, recovery, and relapse among zombie firms by tracking the probability of these events over time. Additionally, the Cox proportional hazards model is employed to assess the impact of firm-specific factors on the likelihood and timing of continuation, recovery, or relapse. As a semi-parametric approach, the Cox model estimates the effect of covariates on the hazard rate, offering insights into the drivers of zombification dynamics. For this analysis, events are defined as follows: Continuation occurs when a firm remains in zombie status in the following year, indicating a persistent zombification spell. Recovery is identified when a firm exits zombie status after being classified as a zombie the previous year, thereby ending a zombification period. Relapse refers to a firm re-entering zombie status after a period of recovery, signalling a return to financial distress. Cases with no relapse by the end of the observation period are treated as right-censored.

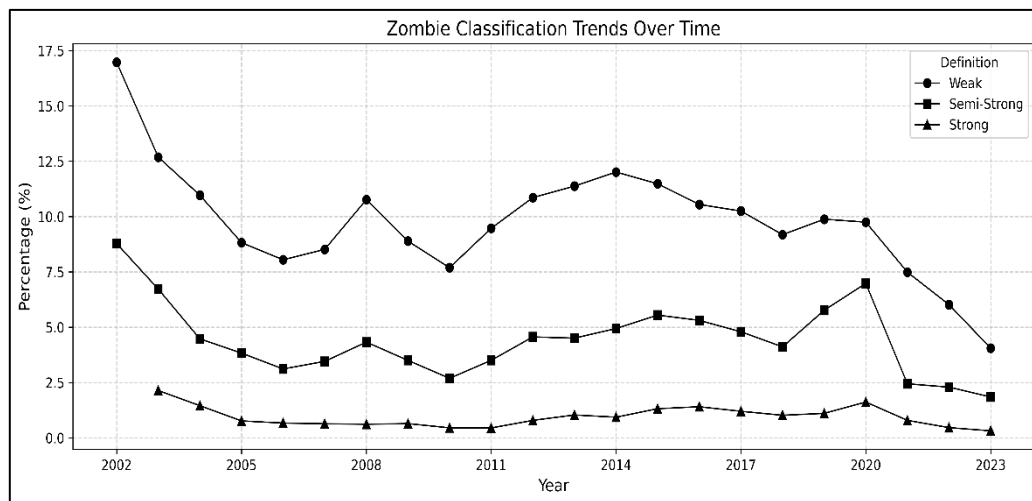
5. RESULTS AND DISCUSSIONS

5.1 Identification of Zombie Firms

Based on the three identification criteria for zombie firms described in the methodology, the trends are depicted in Figure 1.

Share of Zombie firms in India

Figure 1



Note: The share of zombie firms under various definition is calculated as the percentage of zombie firms relative to the total number of firms in a given time period t . Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Considering the weak definition, the trends indicate that, on average over the sample period, approximately 10% of firms in the Indian non-financial sector are classified as zombies. This finding aligns with the results reported by Pattanaik et al. (2022) and Asha Rai et al. (2025). When more stringent conditions are applied, the semi-strong definition identifies, on average, about 4.4% of firms as zombies, whereas the strong definition identifies only about 1%. However, it is important to note that there have been significant variations in these shares across the sample period, 2002- 23. As expected, the share of zombie firms tends to spike during economic downturns, such as the financial crisis of 2008 and the COVID-19-induced recession in 2020, indicating a correlation between economic stress and the prevalence of financially distressed firms. Further, the rise in the share of zombie firms until 2014 and the reversal thereafter coincide with the withdrawal phase of forbearance, during which the RBI, together with the government, began tightening the norms for asset classification (Kulkarni et al., 2021).

Table 2: Industry-wise percentage share of zombie

	Weak Definition	Semi-Strong Definition	Strong Definition
Agriculture and Allied	7.15	3.38	0.59
Construction	7.95	3.93	0.79
IT	9.32	4.01	0.82
Manufacturing	8.23	4.08	0.94
Mining and Quarrying	12.59	5.59	1.34
Others	10.73	4.9	1.21
Public Services	14.82	4.54	0.86
Real Estate and Business	9.79	4.5	0.88
Transport	14.91	4.54	0.8

Note: Companies in the non-financial sector are classified into industry groups based on the three-digit 2008 NIC code.

Having found the overall share of zombie firms in the economy, we next examine their distribution across industry to better understand whether they are concentrated in particular industries. The prevalence of zombie firms varies notably across industries and declines with stricter definitions. Under the weak definition, zombie firms are most prominent in Transport (14.91%), Public Services (14.82%), and Mining and Quarrying (12.59%), while industries like Agriculture and Allied (7.15%), Construction (7.95%), and IT (9.32%) have relatively lower shares. Manufacturing (8.23%), Real Estate and Business (9.79%), and Others (10.73%) lie in the mid-range.

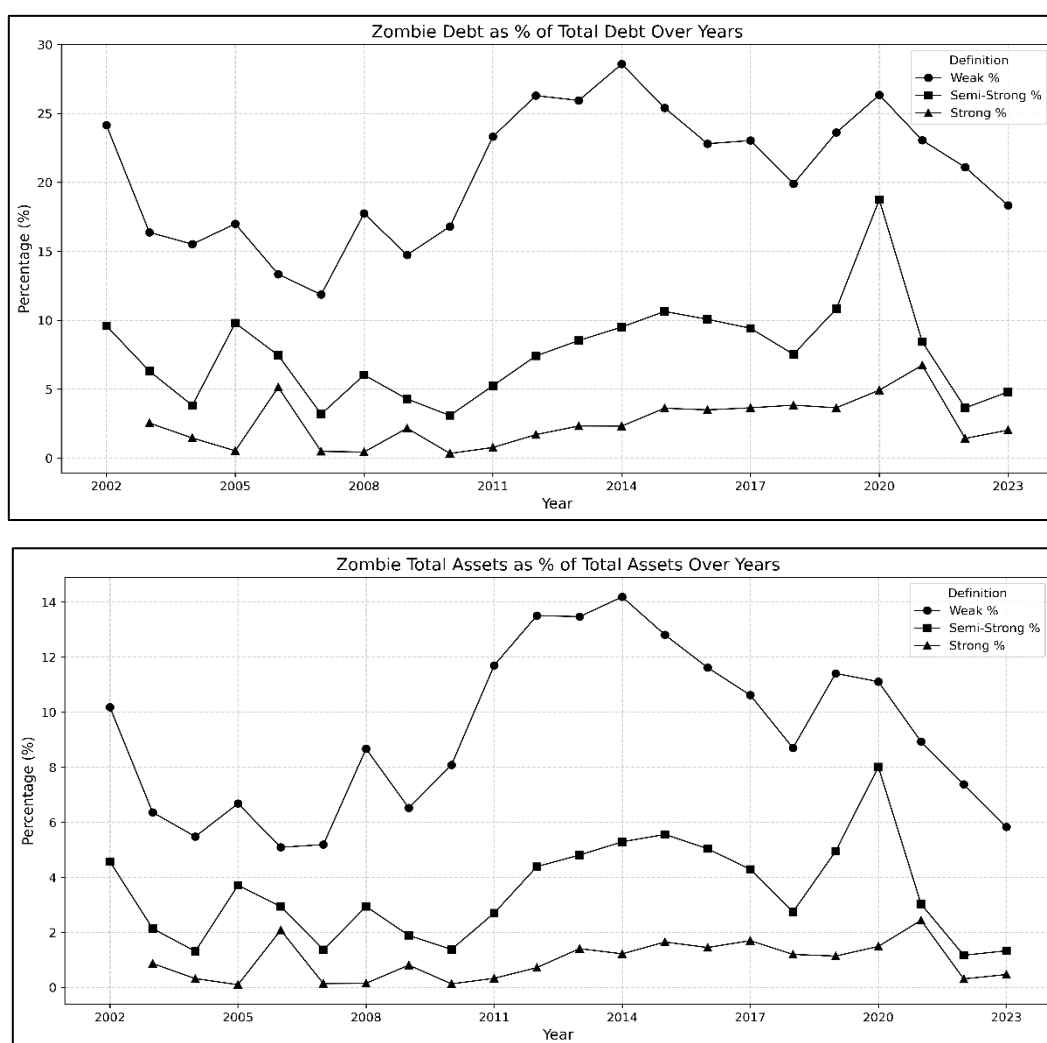
When applying the semi-strong definition, the share reduces across the board, with Mining (5.59%), Others (4.9%), and Agriculture (3.38%) still showing some persistence. Under the strong definition, the zombie share is minimal, with Manufacturing (0.94%), Mining (1.34%), and Others (1.21%) retaining the highest values, while industries such as Agriculture (0.59%), Construction (0.79%), and Transport (0.8%) drop significantly.

Regardless of the definition applied, zombie firms remain highly concentrated in certain industries. Sectors such as Mining and Quarrying, Others, and Manufacturing consistently show higher shares of zombie firms across all three definitions. For instance, under the weak definition, Transport (14.91%), Public Services (14.82%), and Mining (12.59%) report the highest prevalence. Even as the criteria become stricter, these industries continue to feature prominently—Mining (5.59%) and Others (4.9%) under the semi-strong definition, and Mining (1.34%), Others (1.21%), and Manufacturing (0.94%) under the strong definition. In contrast, industries like Agriculture, Construction, and IT consistently show lower zombie presence, with their shares dropping significantly under stricter definitions.

While these results tend to suggest that zombie firms constitute only a small fraction of the overall firm population, a deeper analysis reveals their broader economic implications. Specifically, as shown in Figure 2, zombie firms account for a disproportionately high share of debt and assets under distress. On average, the total debt held by zombie firms, on an average, ranges from 2.4% to nearly 20%, and their share of total assets varies between 1% and 8.8% across the three definitions. These findings underscore that, despite their limited numbers, zombie firms represent a significant concern for the Indian economy.

Share of Total Debt and Total Assets taken over by Zombie

Figure 2



Note: The share of debt and assets held by zombie firms is calculated as the percentage of total firm debt and assets accounted for by zombie firms in a given time period t . Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

5.2 The Structure of Zombie Firms

While zombie firms account for a relatively small share of the overall firm population, their disproportionate share of total debt and assets underscores their economic significance. However, understanding their broader impact also requires a closer look at how these firms differ at the micro level. To this end, this section compares key financial characteristics of zombie and non-zombie firms to uncover patterns in their capital structure, profitability, and investment behaviour.

Table 3: Mean and test of Difference in distribution (Non-Zombie vs Zombie)

Variables	Non-Zombie	Zombie	Kolmogorov-Smirnoff Statistic
Debt Growth ¹	36.25	37.31	0.52***
Borrowing from banks ^{1,3}	17.91	27.57	0.17***
ICR ²	15.27	-4.74	0.81***
ROA ¹	6.92	-1.67	0.35***
Equity share* ^{1,3}	5.27	3.16	0.04***
Inter-corporate loans ^{1,3}	4.16	10.65	0.16***
ROE ¹	4.6	-33.51	0.41***
Capex ^{1,3}	5.36	4.24	0.15***
Loans from promoters, directors and shareholders ^{1,3}	2.14	3.49	0.04***
EDR ²	1.86	-0.11	0.73***
debt to equity ²	2.03	4.72	0.33***
Borrowing from financial institutions ^{1,3}	1.48	3.51	0.07***
Total Assets	5989	6007.1	0.06***
Number of Banks	1.02	1.28	0.03***
Interest Rate Gap	0.94	-0.2	0.05***
Promoter pledging*	0.89	2.1	0.03***
Subsidized Credit ⁴	0.45	0.44	0.01*
Debt to Assets ²	0.39	0.95	0.54***
Net Cashflow ³	0.48	-0.26	0.12***
Altman Z-Score*	1.88	-0.02	0.65***
Tobin's q* ²	1.35	1.21	0.07***
Primary Banks ⁴	0.07	0.18	0.12***
Primary Equity ⁴	0.05	0.01	0.04***
Asset Disposal ^{1,3}	0.18	0.22	0.03***
Borrowings from central & state govt ¹	0.1	0.23	0.02***

Note : * indicates that the variable is only computed for listed firms. *** / ** / * indicate 1%, 5 % and 10 % statistical level of significance respectively. 1 indicates variables are represented in percentages, while 2 indicates ratios, and 3 indicates that variables are scaled by Total Assets. 4 represents a dummy variable. EDR refers to the Earnings to Debt ratio of the firm. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Using the Kolmogorov-Smirnov test, we statistically assess whether these differences in financial metrics are significant between zombie and non-zombie firms (Table 3). Although zombie companies are relatively similar in size, they are less profitable compared to non-zombie firms. Profitability is a significant differentiator between zombie and non-zombie companies: zombie firms exhibit a negative return on assets (ROA) of -1.67%, compared to a positive 6.92% for non-zombie firms.

Additionally, their interest coverage ratio (ICR) is negative, indicating that these firms struggle to generate enough income to cover their interest expenses. Return on equity (ROE) is similarly negative for zombie firms, highlighting their poor financial performance and lack of profitability. Another notable difference is in debt growth, with zombie firms experiencing slightly higher growth (37.31%) than non-zombie firms (36.25%). This suggests that zombie firms are associated with greater dependence on debt financing alongside deteriorating financial performance.

In addition to their higher debt growth, zombie companies have higher debt-to-equity and debt-to-assets ratios. This indicates that zombie firms are more reliant on debt financing and have lower equity participation compared to non-zombie firms. The lower equity share in zombie firms (3.16% vs 5.27% for non-zombie firms) further underscores their reliance on debt rather than equity for financing, consistent with weaker financial conditions among zombie firms.

Investment behaviour also distinguishes zombie companies. They tend to invest less in growth and future opportunities. Capital expenditure (Capex) is lower for zombie firms (4.24%) than for non-zombie firms (5.36%), indicating lower investment activity among zombie firms. Additionally, zombies are more likely to engage in asset disposals, with a value of 0.22% compared to 0.18% for non-zombie firms. This may reflect greater asset disposals among zombie firms relative to non-zombie firms.

The distribution of primary funding sources reveals that zombie firms are significantly more likely to rely on bank financing than on equity, consistent with greater reliance on debt-based funding among financially distressed firms. Zombie companies also exhibit negative cash flows, which is consistent with their overall financial struggles. The Altman Z-Score, a measure of bankruptcy risk, is significantly higher for zombie firms (1.88 vs. -0.02 for non-zombies), further indicating their higher risk of financial distress. In terms of external financing, zombie firms are associated with higher levels of subsidised credit, which may mask the true riskiness

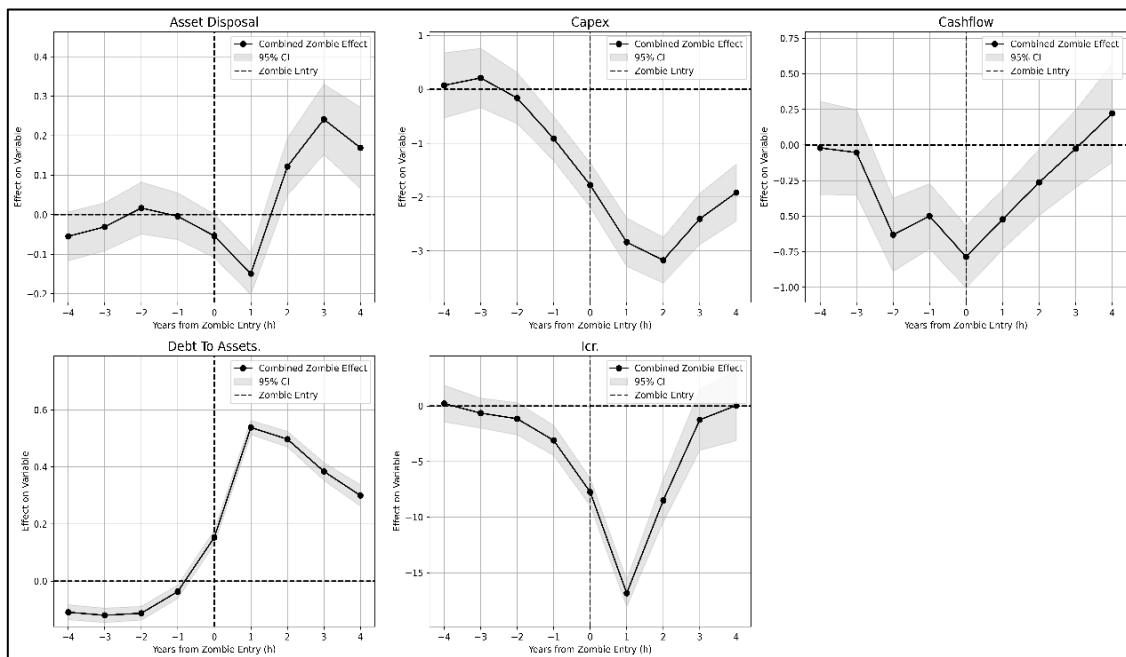
of lending to them, since the interest rate gap between zombie and non-zombie firms is wide and statistically significant, despite the zombies' lower profitability and higher financial risk.

Additionally, zombie firms show a stronger reliance on internal and affiliated financing channels. They receive more inter-corporate loans (10.65% vs 4.16%) and loans from promoters, directors, and shareholders (3.49% vs 2.14%) than non-zombie firms, consistent with greater reliance on internal and affiliated funding sources among zombie firms. Zombie firms also interact with a higher number of banks on average (1.28 vs 1.02) and are more likely to have a primary banking relationship (18% vs 7%), suggesting increased efforts to maintain liquidity and credit access.

5.3 Firm Performance Before and After Zombification

Performance of Key metrics around Zombification

Figure 3



Note: The graph displays impulse response functions estimated using local projection regressions following the method of Jordà (2005). The x-axis represents time horizons (in periods), and the y-axis shows the estimated response of the dependent variable to a one-unit shock in the independent variable. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Borrowing from banks is notably higher among zombie firms (27.57% vs 17.91%), and they also receive more support from central and state government sources (0.23% vs 0.10%). Furthermore, promoter pledging—which is often associated with financial distress—is more common among zombies (2.1% vs 0.89%). Finally, zombie firms exhibit lower market-based valuations, with a lower Tobin's q (1.21 vs 1.35), consistent with weaker market valuations.

To better understand how firm dynamics evolve around zombification over time, we complement the static comparison with a dynamic analysis using local projection regressions. While the preceding section identifies key structural differences between zombie and non-zombie firms at a given point in time, it is equally important to trace how these differences evolve before and after zombification. The local projection analysis (Figure 3) highlights distinct patterns in firm dynamics surrounding zombification. Both capital expenditure and cash flow decline leading up to zombie status. Although capital expenditure shows a modest increase two years after zombification, it remains consistently below the levels observed in non-zombie firms.

In contrast, cash flow begins to rise following zombie entry. This divergence indicates that increases in cash flow among zombie firms are not accompanied by comparable increases in investment activity, suggesting limited improvement in productive investment following zombification. Similarly, ICR also declines significantly before zombification and shows a mild recovery one year after entry, yet it continues to trail far behind the performance of non-zombie firms.

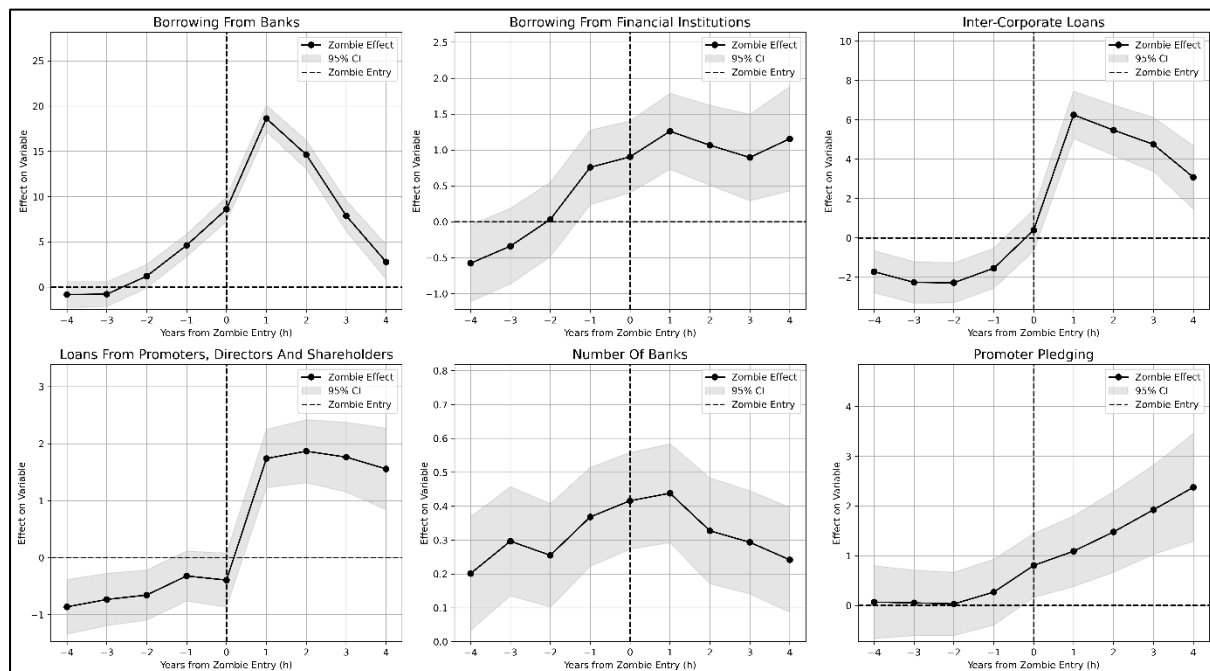
In contrast, Asset Disposal remains below non-zombie levels up to a year after zombification but then rises steadily, staying significantly higher than at non-zombie firms over the period—suggesting relatively higher levels of asset disposal following zombification. Debt-to-Assets, meanwhile, increases steadily throughout the period and then slightly declines after 2 years of zombification, maintaining consistently higher leverage relative to non-zombies. Together, these patterns are consistent with financial deterioration beginning prior to zombification and persisting afterwards, alongside weaker recovery in core performance indicators and continued reliance on debt and asset disposals.

5.4 Financing behaviour of Firms Before and After Zombification

The surge in financing activity around zombie entry needs closer examination. To better understand how firms adjust their financing patterns as they approach financial distress, the local projection regression is extended to trace the dynamics of different funding sources over time. This analysis illustrates how formal and informal sources of finance evolve around zombie entry and whether changes in financing patterns coincide with the transition into zombie status. The results provide a richer picture of the evolution of debt, equity, and internal capital flows surrounding zombie entry, while remaining agnostic about the direction of causality.

Performance of Funding Metrics around Zombification

Figure 4



Note: The graph displays impulse response functions estimated using local projection regressions following the method of Jordà (2005). The x-axis represents time horizons (in periods), and the y-axis shows the estimated response of the dependent variable to a one-unit shock in the independent variable. The share of zombie firms is calculated as the percentage of zombie firms relative to the total number of firms in a given time period t . Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Figure 4 shows a clear rise in different sources of financing before, and at the time firms become zombies, indicating changes in these firms' financing patterns around zombification. Borrowing from banks and financial institutions, and through inter-corporate loans, rises

sharply around zombie entry and remains well above the levels of non-zombie firms, consistent with greater dependence on external credit. Similarly, loans from promoters, directors, and shareholders increase from the period of zombification onward and remain elevated above non-zombie levels, indicating greater reliance on internal and affiliated financing sources.

Promoter pledging starts rising even before the zombie transition and continues to climb afterwards, indicating that promoter pledging is more prevalent around zombification. The number of banks a firm works with also increases slightly before zombie entry and stays significantly higher than non-zombie levels. Overall, the patterns indicate greater reliance on multiple credit sources and higher reliance on debt around zombification. These patterns highlight the importance of examining the association between funding sources and zombie status in greater detail.

5.5 Factors Associated with Zombie Status

Before presenting the regression results, we examine the extent of within-firm variation in the key financing variables, as this informs the choice of the panel estimation strategy. Although financing structures change over time, switching is relatively infrequent, particularly for equity financing. The within-firm standard deviations are 0.201 for primary bank financing and 0.160 for primary equity financing. At the firm-year level, changes occur in 5.4% of observations for primary bank financing and 1.8% for primary equity financing. At the firm level, 24.4% of firms switch their primary bank financing at least once during the sample period, compared with 10.1% for primary equity financing. These findings indicate that financing structures are relatively persistent. Combined with the substantial reduction in sample size under the fixed-effects logit specification, these patterns provide context for our choice of the random-effects model as the baseline estimator. The robustness of the findings is further assessed using alternative specifications.

Having traced how firms evolve around zombification and described the extent of within-firm variation in financing sources, we now examine the association between firm characteristics, funding patterns, and zombie status. A random-effects logistic regression is employed across four model specifications to identify the factors associated with zombie status among India's non-financial firms. Model 1 and Model 2 (Table 4) evaluate the influence of primary financing sources—bank vs. equity—without firm-level controls. Both models show consistent and statistically significant associations: firms primarily financed through banks exhibit higher

odds of being classified as zombie firms (odds ratio ≈ 2.39), whereas firms primarily financed through equity exhibit lower odds of zombie status (odds ratio ≈ 0.31). These associations are robust and consistent across specifications.

In addition, Model 2 includes financial health indicators and firm controls, revealing several significant associations with zombie status. Access to multiple banks (number of banking relationships) and promoter pledging are positively associated with zombie classification. These findings are consistent with the possibility that broader banking networks may be associated with continued access to external financing and that promoter pledging may reflect greater financial vulnerability among distressed firms. However, these mechanisms are not directly tested in the present study. The Group dummy shows a positive and statistically significant relationship with zombie status, contrary to the internal capital market hypothesis, which suggests that business group affiliation should reduce zombification risk. Among financial health controls, higher liquidity (current ratio), profitability (ROA and cash flow), and solvency (EDR) are linked with a lower likelihood of being classified as a zombie, with some effects being small but statistically significant. Larger firm size (log of total assets) also modestly reduces zombie risk, whereas older firms are slightly more prone to zombification. Asset disposals remain statistically insignificant in predicting zombie status.

Finally, Model 3 presents the subsample analysis for listed firms, extending the baseline model by incorporating additional financial health indicators, including the debt-to-equity ratio, return on equity (ROE), and the Altman Z-score. The results indicate that higher leverage, measured by the debt-to-equity ratio, is significantly associated with an increased likelihood of zombie status, consistent with the greater financial risk typically associated with highly leveraged firms. Similarly, a lower Altman Z-score, indicating greater financial distress and bankruptcy risk, is significantly associated with a higher likelihood of zombie status.

In contrast, ROE is not statistically significant in its association with zombie status, suggesting that profitability, as measured by ROE, may be less informative for distinguishing zombie firms in this listed sample. The Group Dummy variable, which captures whether a firm is affiliated with a business group, shows a statistically significant negative association with zombie status in Model 3. This contrasts with earlier findings in Model 2 and implies that, among listed firms, group affiliation is associated with a lower likelihood of zombie status among listed firms, potentially reflecting better access to internal capital markets or support mechanisms within business groups.

Table 4: Results of Panel Logistic Regression

Variables	Model (1)	Model (2)	Model (3)
Primary Banks	0.878*** (0.034)	0.661*** (0.038)	0.094 (0.135)
Primary Equity	-1.155*** (0.095)	-1.205*** (0.101)	-1.54*** (0.158)
Number of Banks		0.041*** (0.006)	-0.002 (0.014)
Promoter pledging		0.006*** (0.001)	0.002 (0.003)
Current ratio		-0.197*** (0.012)	-0.004 (0.036)
Log Total Assets		-0.018** (0.009)	0.095** (0.031)
EDR		-1.411*** (0.029)	-0.629*** (0.113)
Asset Disposal		0.008 (0.016)	-0.055 (0.042)
ROA		-0.025*** (0.001)	0.015** (0.006)
Cashflow		-0.015*** (0.003)	-0.011 (0.010)
Age		0.223*** (0.021)	0.094 (0.076)
Debt to equity			0.017** (0.008)
ROE			-0.001 (0.001)
Altman Z-Score			-1.787*** (0.070)
Group Dummy		0.295*** (0.038)	-0.354*** (0.099)
Constant	-3.919*** (0.118)	-3.643*** (0.149)	-2.575*** (0.436)
Estimator	MLE	MLE	MLE
Firm controls	No	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	264,988	207,793	35,524
Number of firms	23,637	23,146	3,091

Note: Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Other variables, such as promoter pledging and the number of banking relationships, lose significance in this subsample, indicating that their associations with zombie status are less pronounced among listed firms. Liquidity measures, asset disposals, and firm age also remain statistically insignificant, while firm size (log of total assets) surprisingly shows a positive and significant association with zombification in this model, which may reflect differences in financial restructuring or debt management among larger listed firms. In addition, to test the nexus between capital structure and zombification, a set of alternative specifications has been tested, including non-linearity and threshold effects of covariates. All approaches yield consistent results, supporting the robustness of the observed positive association between multiple banking relationships and the likelihood that a firm will be classified as a zombie. In sum, the findings consistently highlight a strong association between financing structure and zombie status. Equity-financed firms are associated with a lower likelihood of zombie classification, whereas bank-dependent firms are associated with a higher likelihood.

5.6 Robustness check

A robustness analysis using alternative definitions of zombie firms is presented in Appendix B.2 to assess the robustness of the main findings. Across both definitions and all model specifications, the core results remain robust. Firms primarily financed by banks consistently exhibit a statistically significant positive association with zombie status, with the coefficient magnitude increasing under the restricted definition. Similarly, reliance on equity financing remains negatively associated with zombie status, reaffirming that equity-funded firms are less likely to be classified as zombies. Furthermore, the number of banking relationships and promoter pledging also maintain a consistent and positive association with zombie status. Overall, the alignment of results across alternative definitions confirms that the principal findings are not sensitive to the choice of zombie definition.

Since the baseline random-effects logit model assumes that the unobserved firm-specific effects are uncorrelated with the explanatory variables, we estimate a correlated random-effects (Mundlak, 1978) specification as a robustness check. This approach relaxes the standard

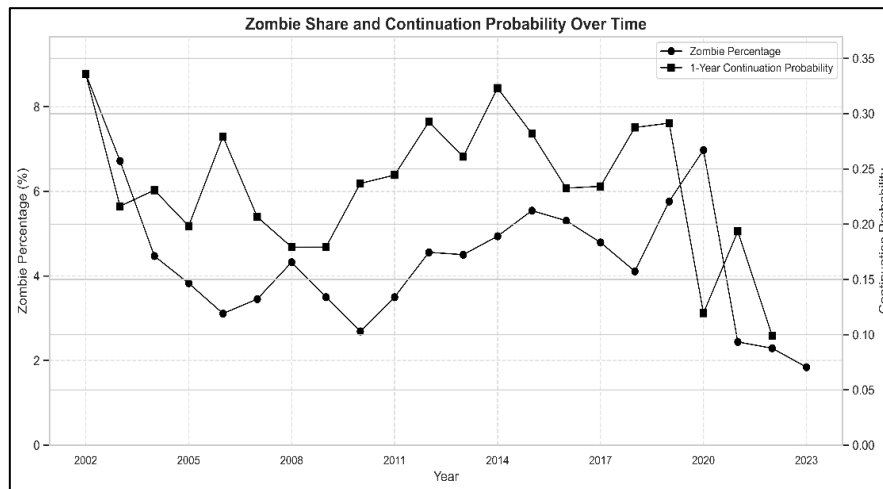
random-effects assumption by including the firm-specific means of the time-varying regressors. The results are broadly consistent with the baseline findings, with the key variables retaining their expected signs and statistical significance in the full-sample specifications. Therefore, the correlated random-effects estimates lead to qualitatively similar conclusions, supporting the robustness of the main findings.³

We also conducted an additional robustness check by applying a stricter maturity filter and redefining zombie firms to require a minimum firm age of 10 years rather than 5 years. The results remain qualitatively similar to the baseline specification. Although the stricter filter reduces the number of firms classified as zombies, the overall time-series pattern and cyclical dynamics of zombie prevalence remain largely unchanged. As shown in Appendix B.1, the zombie percentages under the 10-year filter closely mirror those obtained under the baseline 5-year definition, suggesting that the study's conclusions are not sensitive to the choice of maturity threshold.

5.7 Life Cycle of Zombie Firms

In contrast to most existing studies, which focus predominantly on the incidence of zombification, this section and the following section advance the analysis by examining the lifecycle progression of zombie firms. In analysing the persistence of zombie firms, Figure 5 shows that the probability of a firm remaining a zombie from one year to the next has ranged from 10% to 35% over the observed period. This continuation probability, along with the overall share of zombie firms, increased notably after 2008 and peaked around 2014, coinciding with a period of regulatory forbearance in India. In response, both the Government of India and the Reserve Bank of India introduced stricter guidelines on asset classification and income recognition. A significant policy intervention during this time was the enactment of the Insolvency and Bankruptcy Code (IBC) in 2016, which aimed to improve the resolution of corporate distress and curb the survival of non-viable firms. The decline in zombie continuation rates observed in subsequent years is consistent with the timing of these regulatory and institutional reforms. These trends highlight the importance of further examining the persistence of zombie firms and the factors associated with their continued status.

³ Detailed results from the correlated random-effects (Mundlak, 1978) robustness analysis are available from the authors upon request.



Note: The share of zombie firms is calculated as the percentage of zombie firms relative to the total number of firms in a given time period t . The probability of a zombie firm persisting for one year is calculated by dividing the number of firms that remain zombies in period $t+1$ by the total number of zombie firms in period t . Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

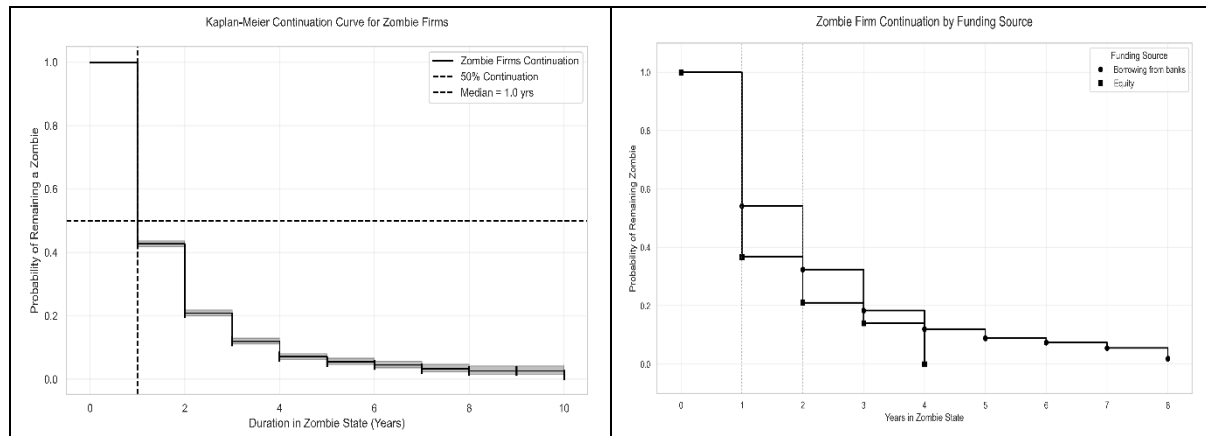
To better understand these dynamics, it is essential to go beyond aggregate trends and examine how long firms remain in zombie status. This motivates a survival analysis to explore the persistence of zombification and how it varies by funding source. The Kaplan Meier survival analysis in Figure 6 (left panel) illustrates that fewer than half of zombie firms remain in zombie status after one year, and less than a quarter persist beyond two years.

While zombie persistence varies across funding sources, survival also varies sharply across industries. Breaking down zombie persistence by sector highlights differences in recovery and persistence patterns across firms. An industry-level segregation (Figure 7) reveals that zombie firms in sectors such as Information Technology, Public Services, Real Estate and Business, and Transport generally remain in zombie status for less than five years.

5.7.1 Survival of Zombie Firms

Survival of zombie firms

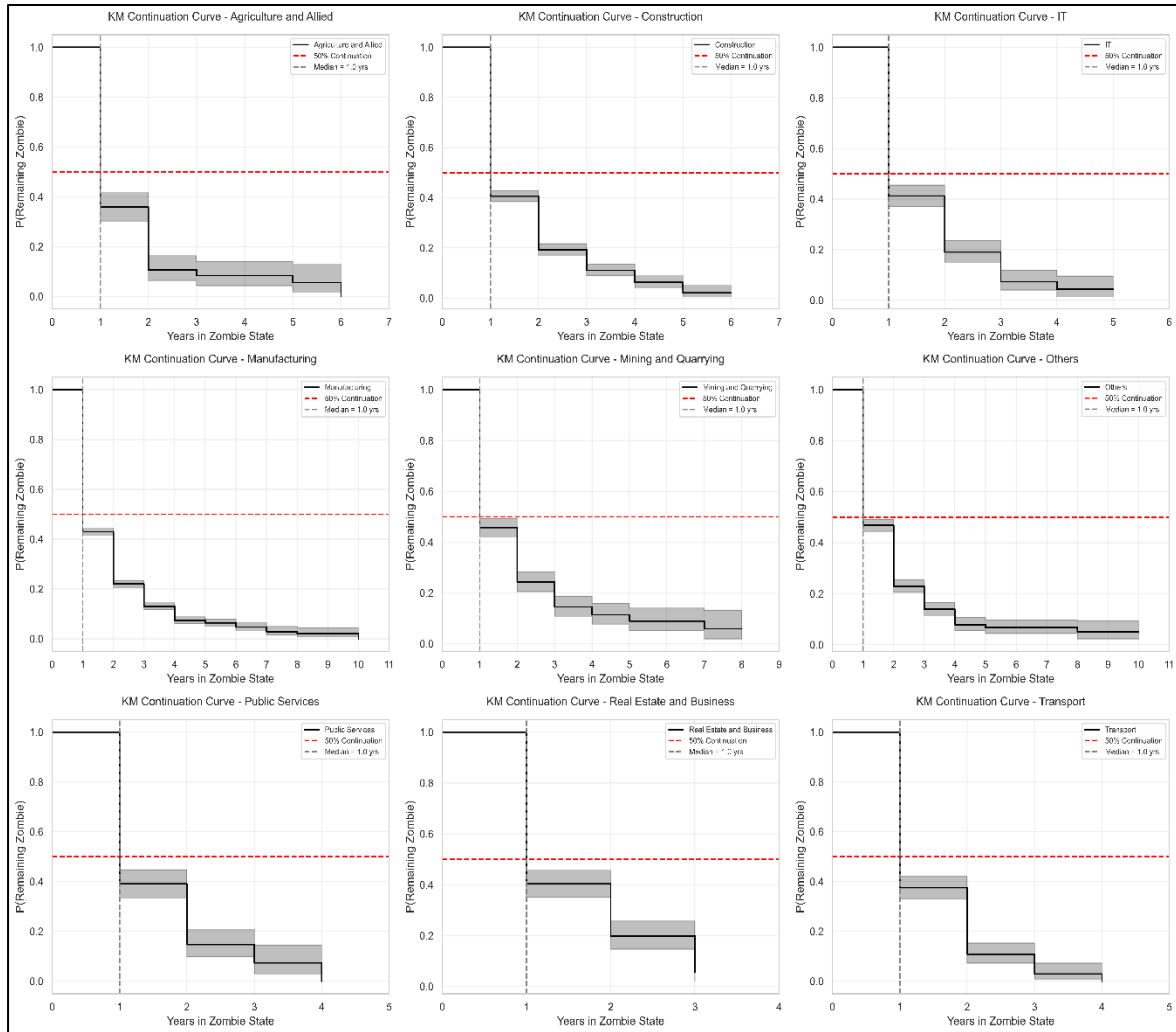
Figure 6



Note: The survival function indicates the probability that a firm classified as a zombie in period t continues to remain a zombie in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The funding source is determined based on the criterion that it must constitute more than 50% of the firm's total assets. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

When segregated by primary funding source, the survival patterns differ significantly. Figure 6 (right panel) shows that less than 40% of zombie firms primarily funded through equity remain zombies after one year, compared to more than 50% of those funded by banks. Further, zombie firms primarily funded through equity are associated with substantially lower persistence beyond the fourth year, whereas those primarily funded through banks are associated with longer persistence, extending up to eight years. Furthermore, the log-rank test (Appendix C) statistically indicates a significant difference in the duration of zombie status between equity-funded and bank-funded firms.

In contrast, firms in industries like Agriculture, Construction, Manufacturing, Mining and Quarrying, and other sectors tend to persist in zombie status for more than five years. This pattern is consistent with earlier findings in the study, which indicated that these industries not only have a higher prevalence of zombie firms but also exhibit greater persistence over time.



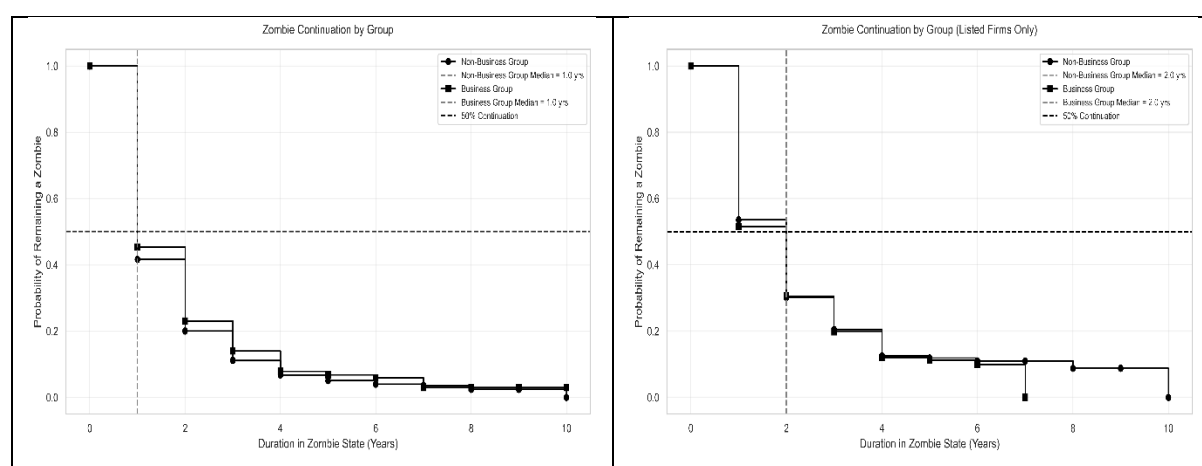
Note: The survival function indicates the probability that a firm classified as a zombie in period t continues to remain a zombie in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The classification into broad industries is based on the three-digit National Industrial Classification (NIC) code. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Finally, Figure 8 shows that firms affiliated with the business group have a higher probability of continuing compared to firms that are not affiliated with any business group. Given the contrasting results observed in the logistic regression, a subsample Kaplan-Meier survival analysis was conducted exclusively for listed firms. This analysis is consistent with the regression findings and contrasts with the general trend observed among business group-affiliated firms.

While the broader sample suggests that firms affiliated with business groups have a higher probability of remaining in zombie status over time, the Kaplan-Meier results for listed firms indicate that those with business group associations exhibit a slightly lower probability of continuation in zombie status compared to firms without such affiliations.

Zombie Firm Continuation by Business Group Affiliation

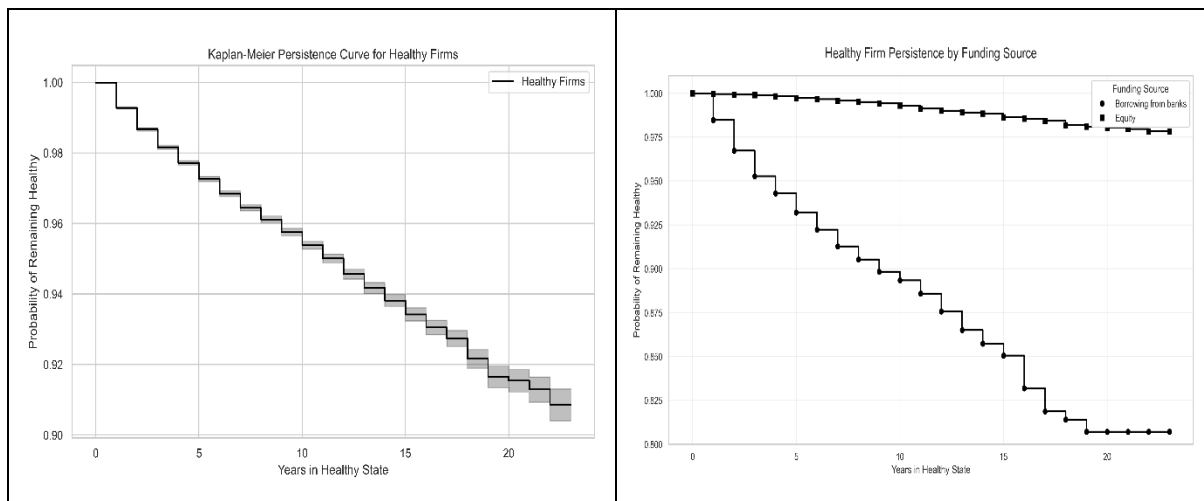
Figure 8



Note: The survival function indicates the probability that a firm classified as a zombie in period t continues to remain a zombie in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The survival analysis distinguishes firms using a group dummy variable, which indicates whether a firm is part of a business group. Additionally, a subsample analysis is conducted exclusively for listed firms, identified by a separate variable that denotes listing status. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

To provide a comprehensive understanding of firm dynamics, it is essential not only to analyze the persistence of zombie status but also to examine the survival and continuation patterns of healthy firms within their non-zombie state. Conducting Kaplan-Meier survival analysis on healthy firms enables a comparative perspective on the duration of firms maintaining financial health. This approach helps to contextualise the persistence of zombie status by contrasting it with the stability of healthy firms, thereby offering insights into the factors associated with firm resilience and longevity across different financial conditions.

A complementary analysis was conducted (Figure 9) to examine the persistence of non-zombie firms, and the results shows that their probability of transitioning into zombie status remained relatively low, rising to only about 10% over the entire study period (left panel).



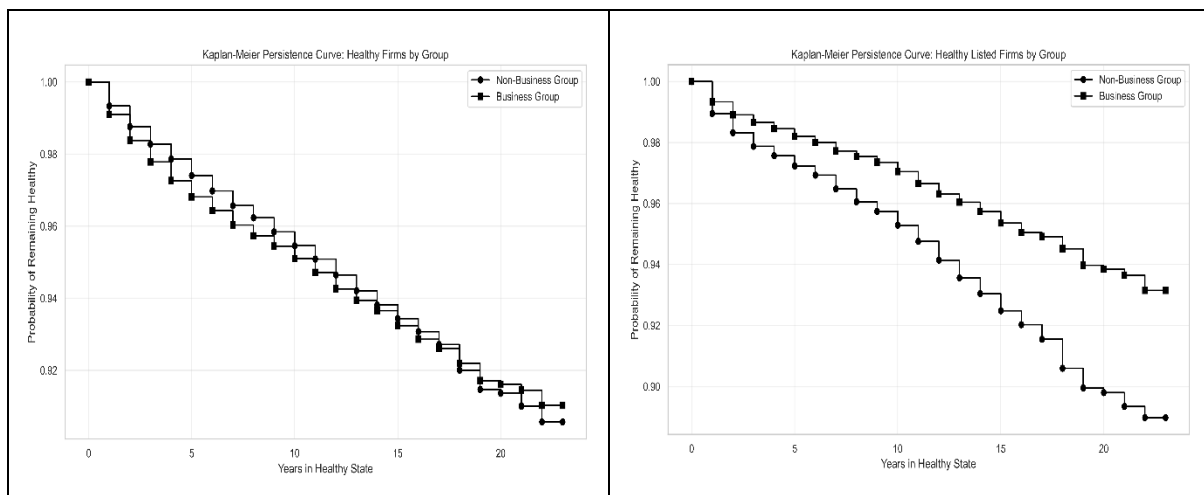
Note: The survival function represents the probability that a firm classified as healthy in period t continues to remain healthy in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until entry into zombie status. The funding source is determined based on the criterion that it must constitute more than 50% of the firm's total assets. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Disaggregating by primary funding source further reveals (right panel) that more than 97.5% of non-zombie firms primarily funded through equity remained non-zombies throughout the sample period, whereas only around 80% of those primarily funded through banks remained non-zombies over time. These differences indicate an association between primary funding source and the persistence of non-zombie status, without implying a causal relationship.

In addition, Figure 10 indicates that healthy firms without any business group affiliation generally have a marginally higher probability of remaining in a healthy state over time. Nonetheless, the probability of remaining a healthy firm throughout the study period remains above 90%, irrespective of business group affiliation. The results from the subsample analysis of listed firms remain consistent, showing that listed firms affiliated with business groups have a higher probability of continuing as healthy firms compared to those without any business group affiliation.

Healthy Persistence by Business Group Affiliation

Figure 10

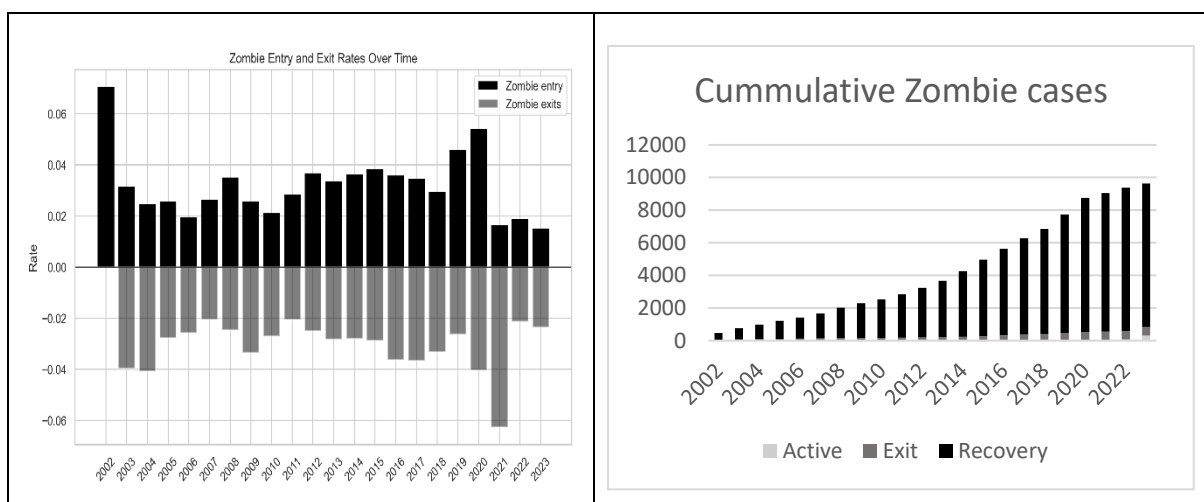


Note: The survival function represents the probability that a firm classified as healthy in period t continues to remain healthy in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until entry into zombie status. The survival analysis distinguishes firms using a group dummy variable, which indicates whether a firm is part of a business group. Additionally, a subsample analysis is conducted exclusively for listed firms, identified by a separate variable that denotes listing status. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

5.7.2 Recovery of Firms from Zombie Status

Zombie Entry, Exits and Cumulative Zombie cases

Figure 11



Note: Entry into zombie status is defined as the first transition from a non-zombie to a zombie state. A zombie spell continues for as long as the firm remains classified as a zombie. Recovery refers to a transition from zombie to non-zombie status, while relapse denotes a re-entry into zombie status following a prior recovery episode. Exit refers to the missing information in the database till the end of the sample period. Specifically, we didn't cross-check the status of the firm to verify the exact cause of exits. For all practical purposes we assumed that absence in the database as exit. Zombie firms are identified using the semi-strong definition.

While most non-zombie firms stay healthy over time, the story is different for zombies. The next analysis shifts focus to how zombie firms exit—whether by recovery or otherwise. Figure 11 (left panel) shows that fluctuations in the number of zombie firms reflect both new entrants and exits, measured as a percentage of total firms observed in each year, measured as a percentage of total firms observed in each year. However, the analysis (right panel) shows that the majority of zombie firm exits are associated with recovery, with a significant proportion of firms exiting zombie status through recovery.

Table 5: Zombie Entry, recovery and Exit stats

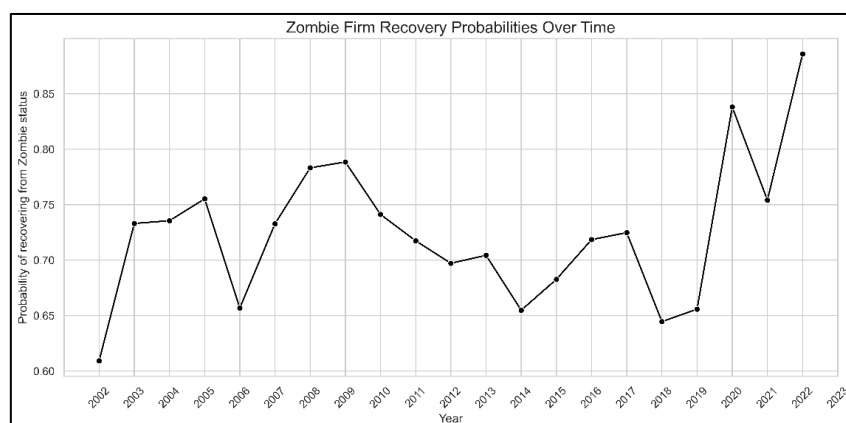
Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Zombie Entry	476	296	208	237	204	247	346	279	228	315	393
Recovery	436	275	194	222	184	229	328	268	221	299	384
Exit	40	21	14	15	20	18	18	11	7	16	9

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Zombie Entry	433	594	706	675	636	574	892	1013	305	333	250
Recovery	413	570	664	627	594	523	829	964	278	295	0
Exit	20	23	42	47	41	51	59	47	16	5	0

Table 5 reports the annual number of firms entering zombie status along with their subsequent recovery and exit outcomes over the sample period. For instance, among the 476 firms that entered zombie status in 2002, 436 firms eventually recovered in subsequent years, while 40 firms exited without an observed recovery. Similar patterns are observed across later entry cohorts, indicating that although a substantial proportion of zombie firms recover over time, a smaller share remain unresolved until the end of the sample period. To provide a more detailed understanding of the transition dynamics and timing of recovery and exit outcomes, the corresponding transition matrices are reported in Appendix D.

Recovery Probability of Zombie Firms over time

Figure 12

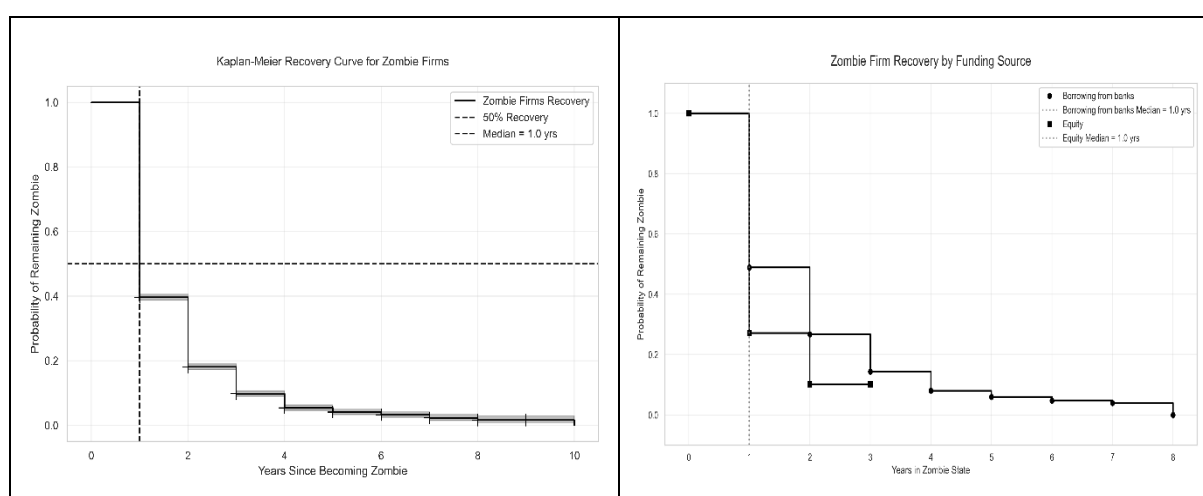


Note: The probability of a zombie firm recovering for one year is calculated by dividing the number of firms that recover from zombie status in period $t+1$ by the total number of zombie firms in period t . Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Further, Figure 12 shows that the one-year recovery probability of zombie firms has increased steadily in recent years, indicating an improvement in recovery outcomes over the sample period. To provide a deeper understanding of the recovery process, the Kaplan–Meier estimates of zombie firm recovery are presented in the following section.

Recovery of Zombie Firms

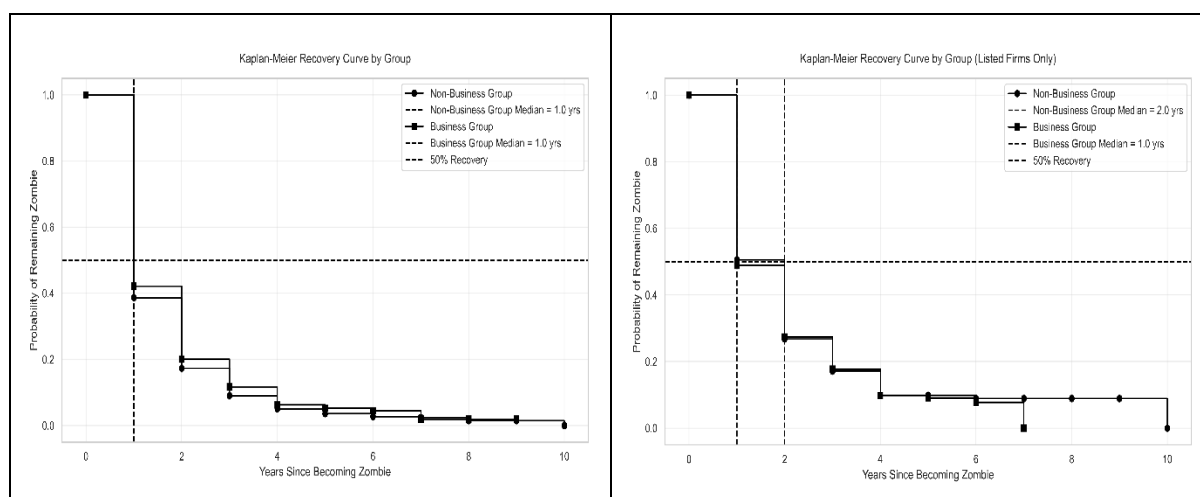
Figure 13



Note: The survival function indicates the probability that a firm classified as a zombie in period t recovered from zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The funding source is determined based on the criterion that it must constitute more than 50% of the firm's total assets. Zombie firms are defined using semi-strong zombie definition.

A similar survival analysis was carried out to examine the recovery patterns of zombie firms. Figure 13 (left panel) shows that over 50% of zombie firms recover within one year of zombification, and more than 90% recover within three years. Consistent with previous findings, Figure 13 (Right panel) shows that the bank-funded firms are associated with slower recovery patterns, with only about 50% of zombie firms recovering after one year.

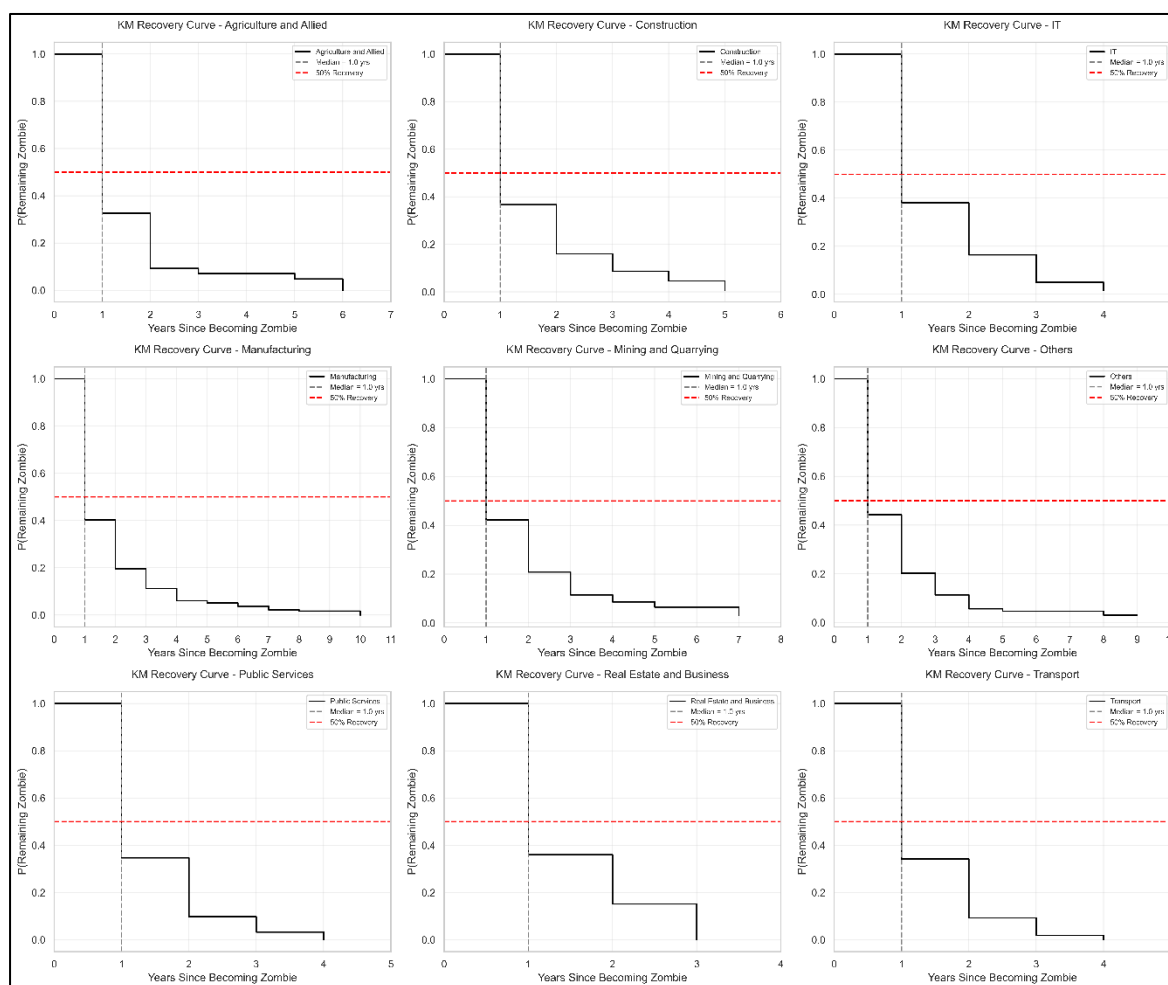
In contrast, over 70% of zombie firms primarily funded through equity recover within the same time frame. This suggests a significant association between funding source and the recovery dynamics of zombie firms, a pattern further supported by the statistically significant results of the log-rank test (Appendix C).



Note: The survival function indicates the probability that a firm classified as a zombie in period t recovered from zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The survival analysis distinguishes firms using a group dummy variable, which indicates whether a firm is part of a business group. Additionally, a subsample analysis is conducted exclusively for listed firms, identified by a separate variable that denotes listing status. Zombie firms are defined using semi-strong definition. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Building further on the factors associated with recovery, Figure 14 reveals that business group affiliation is associated with relatively slower recovery from zombie status, whereas firms without business group affiliation exhibit relatively faster recovery patterns in comparison to those affiliated with a business group. Although the difference between firms with and without business group affiliation is not substantial, the results for the listed-firm subsample remain consistent with the regression findings: the median time to recover from zombie status is 2 years for firms without business group affiliation, compared to 1 year for firms affiliated with a business group.

Finally, Figure 15 shows that firms in industries such as Construction, Information Technology, Public Services, Real Estate and Business, and Transport tend to recover from zombie status within five years. This observation aligns with earlier findings where these industries exhibited a relatively lower share of zombie firms. Conversely, sectors like Agriculture and Allied, Manufacturing, Mining and Quarrying, and Other Industries—where the share of zombie firms was found to be higher—typically take more than five years to recover, consistent with relatively greater persistence of zombie status in these sectors.



Note: The survival function indicates the probability that a firm classified as a zombie in period t recovered from zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The classification into broad industries is based on the three-digit National Industrial Classification (NIC) code. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Therefore, the high recovery rate of zombie firms raises the question of whether the concept of a 'zombie firm' is more a transient phase rather than a true indicator of firm viability. To answer this, the study compares the characteristics of recovered zombies with those of firms that have never been zombies. Focusing on recovered zombie firms, the analysis suggests that although most zombie firms recover, they continue to exhibit several characteristics commonly associated with zombie firms.

Table 6: Mean and test of difference in distribution (Recovered vs Never-been zombie)

Variables	Never-been Zombie	Recovered Zombie	Kolmogorov- Smirnov Statistic
Total Assets	6106.77	5688.21	0.04***
Debt Growth ¹	39.42	28.24	0.09***
ICR ²	18.63	3.58	0.35***
Borrowing from banks ^{1,3}	16.5	23	0.12***
ROA ¹	8.09	2.61	0.24***
ROE ¹	8.35	-12.76	0.25***
Equity* ^{1,3}	5.5	1.36	0.03***
Capex ^{1,3}	5.38	5.13	0.08***
Inter-corporate loans ^{1,3}	3.00	8.13	0.13***
EDR ²	2.29	0.46	0.35***
Loans from promoters, directors and shareholders ^{1,3}	1.92	2.9	0.03***
Debt to equity ²	1.66	3.57	0.24***
Interest Gap	1.26	-0.07	0.07***
Borrowing from financial institutions ^{1,3}	1.16	2.59	0.06***
Number of Banks	0.96	1.19	0.05***
Promoter pledging*	0.68	1.62	0.03***
Subsidized Credit ⁴	0.44	0.49	0.05***
Net Cashflow ³	0.55	0.20	0.06***
Debt to Assets ²	0.32	0.64	0.33***
Altman Z-Score*	2.13	1.21	0.29***
Tobin's q* ²	1.43	1.20	0.07***
Asset Disposal ³	0.16	0.24	0.03***
Primary banks ⁴	0.05	0.13	0.08***
Primary Equity ⁴	0.05	0.03	0.02***

Note : * indicates that the variable is only computed for listed firms. *** / ** / * indicate 1%, 5 % and 10 % statistical level of significance respectively. 1 indicates variables are represented in percentages, while 2 indicates ratios, and 3 indicates that variables are scaled by Total Assets. 4 represents a dummy variable. EDR refers to the Earnings to Debt ratio of the firm. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

The Kolmogorov-Smirnov statistic (Table 6) reveals a significant difference in distribution between recovered zombie firms and never-been-zombie firms, particularly in important financial metrics. For instance, the Interest Coverage Ratio (ICR) shows a substantial difference, with recovered zombie firms struggling to generate sufficient income to meet their interest obligations (ICR of 3.46 times) compared to never-been-zombie firms (ICR of 18.9 times).

This is consistent with weaker debt-servicing capacity among recovered zombie firms. Similarly, the Return on Equity (ROE) for recovered zombie firms is negative (-12.54%), while never-been-zombie firms achieve a positive return (8.56%), consistent with weaker profitability among recovered zombie firms. Further, recovered zombie firms lag in debt growth when compared with never-been-zombie firms, exhibiting higher debt growth (39.42%) compared to the relatively lower (28.24%) of recovered zombie firms. This slower pace of debt accumulation may reflect differences in financing patterns between recovered zombie firms and never-been-zombie firms.

Furthermore, recovered zombie firms show a significantly higher debt-to-assets ratio (0.64 times) compared to never-been-zombie firms (0.32 times), indicating relatively higher debt dependence among recovered zombie firms. The debt-to-equity ratio also is consistent with relatively higher financial risk among recovered zombie firms, with a ratio of 3.57, much higher than the 1.66 of never been zombie firms.

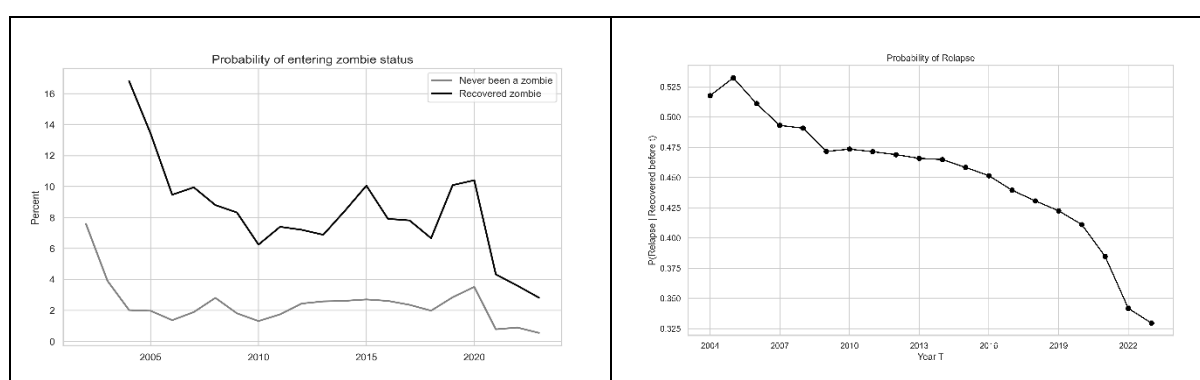
This is consistent with greater reliance on borrowed funds relative to equity financing among recovered zombie firms. In addition, recovered zombie firms engage with more banks (1.19 compared to 0.96 for never-been-zombie firms), which may reflect greater dependence on multiple financing relationships. Finally, promoter pledging in recovered zombie firms is higher (1.62%) compared to never-been-zombie firms (0.68%), suggesting greater use of pledged promoter shares among recovered zombie firms. This pattern is consistent with relatively weaker financial conditions and continued reliance on external financing among recovered zombie firms. Beyond profitability and debt ratios, several other dimensions reveal that recovered zombie firms continue to lag behind healthier peers. Despite slightly smaller average total assets (5688.21 vs 6106.77), they rely more heavily on bank borrowing (23% vs 16.5%) and inter-corporate loans (8.13% vs 3%), suggesting relatively greater reliance on external debt financing channels.

They also exhibit lower equity shareholdings (1.36% vs 5.5%) and higher loans from promoters and directors (2.9% vs 1.92%), suggesting relatively greater dependence on affiliated financing sources. Profitability indicators like ROA (2.61% vs 8.09%) and EDR (0.46 vs 2.29) remain significantly lower for recovered zombies. Furthermore, their Altman Z-Score (1.21 vs 2.13) and Tobin's q (1.20 vs 1.43) are consistent with relatively weaker financial fundamentals and market valuations.

Recovered zombies also face tougher financing conditions, with a negative interest rate gap (-0.07 vs 1.26), suggesting relatively less favourable borrowing conditions compared to healthier firms. Their net cash flows are weaker (0.20 vs 0.55), and they receive marginally more subsidized credit (0.49 vs 0.44). The data further reveal more frequent asset disposals (0.24 vs 0.16), a higher likelihood of financing from primary banks (0.13 vs 0.05), and slightly lower use of primary equity (0.03 vs 0.05), all of which are consistent with continued dependence on external debt financing among recovered zombie firms.

Probability of entering into Zombie Status

Figure 16



Note: The probability of relapse is calculated by dividing the number of firms that revert to zombie status in period $t+1$, given that they had recovered in a prior period $t < T$. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Overall, these significant differences across important financial metrics highlight that recovered zombie firms continue to exhibit weaker financial performance, compared to firms that have never been classified as zombies, with weaker profitability, higher debt reliance, and a lower income-generating capacity. These patterns suggest that recovered zombie firms may remain financially fragile despite exiting zombie status. As a result, the study further examines the potential for relapse among recovered zombie firms. The lifecycle analysis suggests that recovery from zombie status does not necessarily correspond to a full restoration of financial health. Moreover, Figure 16 shows that the probability of entering zombie status⁴ is significantly higher for recovered zombies compared to never-been zombie firms.

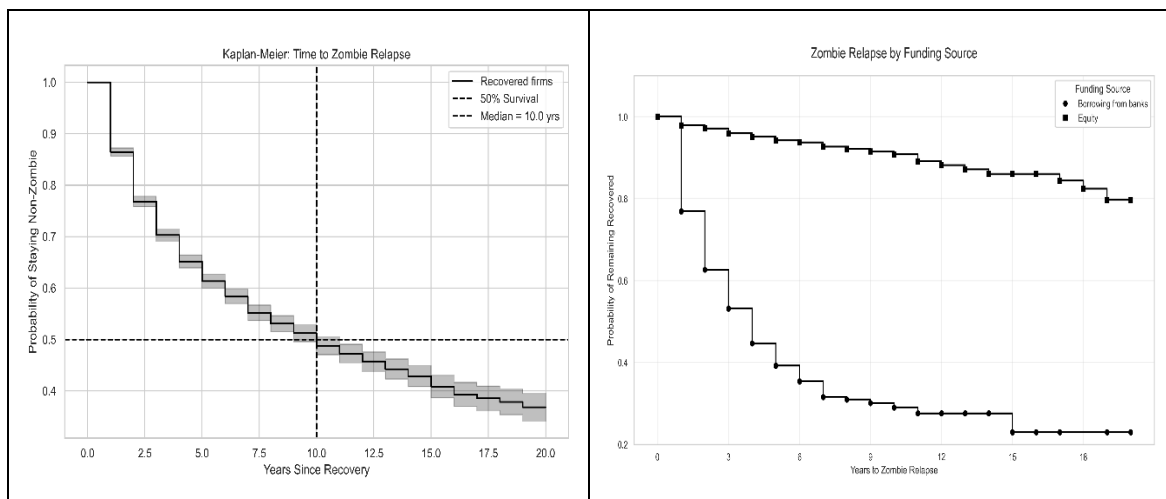
⁴For the calculation of the probability of entering zombie status, see Banerjee & Hofmann (2022, p. 20)

5.7.3 Relapse of Recovered Zombie Firms

Consistent with the findings of Banerjee & Hofmann (2022), we also observe that recovered zombie firms exhibit a tendency to relapse into zombie status within a few years. The Kaplan-Meier plots (Figure 17) show that over half of recovered zombie firms relapse within 10 years. Consistent with previous survival estimates, only about 20% of recovered equity-funded firms relapse, while nearly 80% of recovered bank-funded firms relapse into zombie status. The log-rank test (Appendix C) indicates a statistically significant difference in relapse rates between bank and equity-funded recovered zombie firms.

Relapse of Recovered Zombie

Figure 17



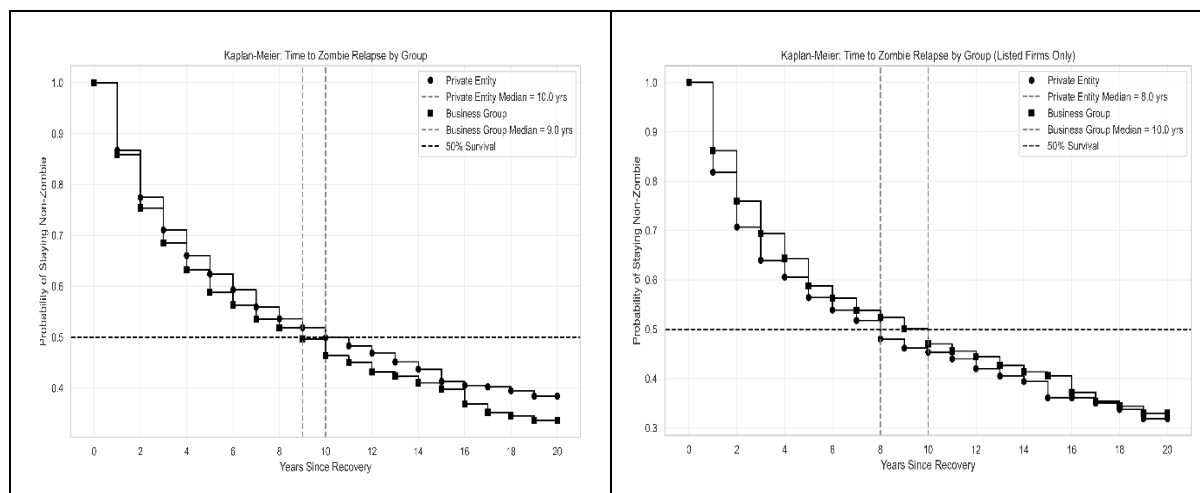
Note: The survival function indicates the probability that a firm classified as a recovered in period t relapsed into zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The funding source is determined based on the criterion that it must constitute more than 50% of the firm's total assets. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Extending the analysis to beyond recovery, Figure 18 explores the likelihood of firms falling back into zombie status. Nearly 50% of recovered firms affiliated with a business group relapse within nine years, while half of the non-affiliated firms relapse within ten years. Contrary to the general trend but in line with regression results for listed firms, the subsample Kaplan-Meier plot shows that among listed firms, those without group affiliation exhibit relatively faster relapse pattern, than those with business group affiliation.

Finally, Figure 19 shows that in nearly all industries, half of the recovered firms relapse into zombie status within ten years of recovery. Moreover, in every industry, more than 60% of the recovered zombie firms eventually relapse into zombie status.

Relapse of Zombie firms by Business Group Affiliation

Figure 18

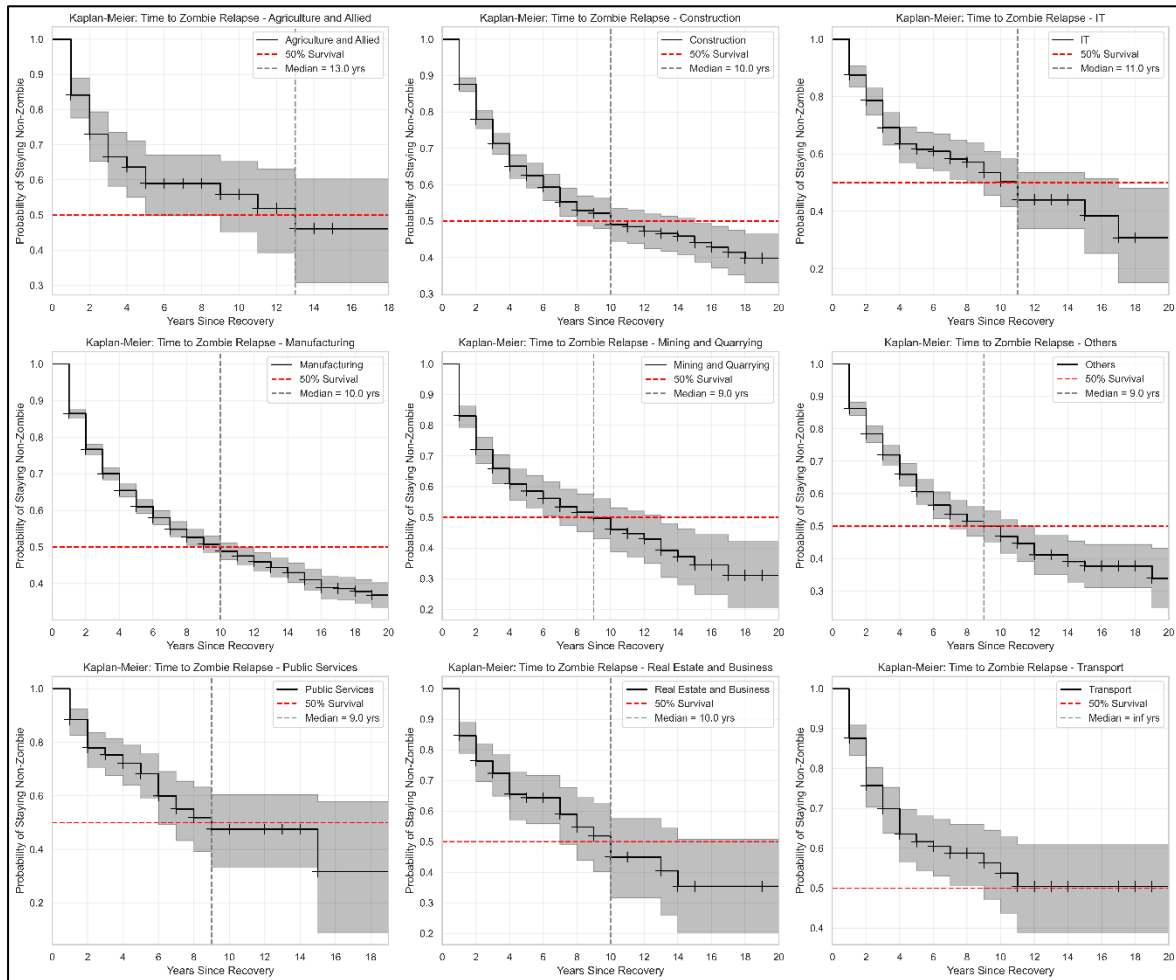


Note: The survival function indicates the probability that a firm classified as a recovered in period t relapsed into zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The survival analysis distinguishes firms using a group dummy variable, which indicates whether a firm is part of a business group. Additionally, a subsample analysis is conducted exclusively for listed firms, identified by a separate variable that denotes listing status. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

5.7.4 Cox Proportional Hazard Model

While the non-parametric Kaplan-Meier analysis highlights notable differences across funding sources, we deploy a semi-parametric Cox Proportional Hazards model to precisely estimate the relationship between funding types and other factors affecting zombie continuation, recovery and relapse.

The results of the Cox Proportional Hazards model (Table 7) show significant relationships between funding sources and the likelihood of firms' recovery, continuation, and relapse. Equity funding is associated with a decrease in the likelihood of a firm relapsing into zombie status, indicating that equity-funded firms are significantly less likely to fall back into zombie conditions.



Note: The survival function indicates the probability that a firm classified as a recovered in period t relapsed into zombie status in subsequent periods. Censoring is accounted for, and the estimates reflect the duration until exit from zombie status. The classification into broad industries is based on the three-digit National Industrial Classification (NIC) code. Zombie firms are defined as those that simultaneously meet all of the following conditions over a given period: an interest coverage ratio (ICR) below 1, leverage above the industry median, positive debt growth, a net profit margin below the industry median, and negative sales growth. To be declassified as a zombie firm, a company must no longer meet one or more of these criteria over a sustained period.

Table 7: Cox Proportional Hazard model results

Variables	Continuation	Recovery	Relapse
Primary Bank	0.67*** (0.05)	-0.23*** (0.04)	0.65*** (0.05)
Primary Equity	-0.46 (0.33)	0.32 (0.19)	-2.07*** (0.18)
Number of Banks	0.02** (0.01)	-0.03*** (0.01)	0.01*** (0.01)
Subsidized Credit	0.05 (0.04)	0.05 (0.03)	0.16*** (0.04)
Promoter Pledging	0.01*** (0.00)	-0.01*** (0.00)	-0.00* (0.00)
Group Dummy	0.02 (0.06)	-0.05 (0.04)	0.21*** (0.05)
Listed firm	0.11 (0.07)	-0.22*** (0.06)	0.43*** (0.07)
Group dummy * Listed firm	-0.32*** (0.10)	0.12 (0.09)	-0.25** (0.10)
Estimator	MLE	MLE	MLE
Observations	11,424	10844	8,273
p-value (χ^2 test)	0.00025	0.00034	0.10423

Note : *** p < 0.01, ** p < 0.05, * p < 0.1. Proportional-hazards assumption is tested based on Schoenfeld residuals.

This suggests that equity funding is associated with lower relapse probability among recovered zombie firms. In contrast, bank-funded firms exhibit a higher likelihood of relapse, suggesting relatively higher relapse risk among bank-funded firms. Moreover, bank funding is associated with a decrease in the likelihood of recovery from zombie status, suggesting that bank-funded firms are associated with a lower likelihood of recovery from zombie status. Additionally, the number of banks involved in a firm's funding is consistently associated with an increased likelihood of relapse, continuation, and a reduced likelihood of recovery from zombie status. Specifically, each additional banking relationship is associated with approximately a 3% higher relapse likelihood, along with lower recovery likelihood and higher continuation probability.

Subsidised credit, while increasing the likelihood of relapse, has little impact on recovery and continuation, though its effect on relapse is statistically significant. Promoter pledging is significantly associated with all three outcomes considered in the model (continuation, recovery, relapse). Being a listed firm is associated with a 19.8% lower hazard of recovery,

indicating that listed zombie firms exhibit relatively lower recovery hazards. At the same time, listed firms face a 53.7% higher hazard of relapse, suggesting they are significantly more prone to falling back into zombie status once recovered. However, the interaction between listing and group affiliation reveals important nuances. Group-affiliated listed firms have a 27.4% lower hazard of continuing as zombies and a 22.1% lower hazard of relapse. The underlying mechanisms, including potential access to internal capital markets or strategic support within business groups, are not directly examined in the present study. This suggests that group affiliation is associated with relatively lower relapse hazards among listed firms. The underlying mechanisms, including potential access to internal capital markets or strategic support within business groups, are not directly examined in the present study.

Business group affiliation is associated with shorter recovery and continuation durations and a higher likelihood of relapse. These results highlight significant associations between funding structure and zombie-firm dynamics, particularly with respect to equity and bank funding. The robust χ^2 test p-values for all models support the statistical significance of these findings, and the Schoenfeld residuals support the proportional hazards assumption for the model.

These lifecycle patterns also carry broader implications for credit allocation and restructuring efficiency in India. The longer persistence and higher relapse rates observed among bank-funded zombie firms may reflect differences in the financing patterns of distressed firms. However, the present analysis does not directly identify the mechanisms underlying these associations, including the potential roles of delayed restructuring or continued credit rollover. In contrast, the relatively faster recovery observed among equity-funded firms is consistent with an association between financing structure and the durability of firm recovery. These findings are particularly relevant in the Indian context, where the banking sector plays a dominant role in corporate financing and where recent reforms such as the Insolvency and Bankruptcy Code (IBC) have aimed to improve the resolution of distressed firms and strengthen credit discipline.

6. CONCLUSION

This study provides a comprehensive analysis of zombie firms in India's non-financial corporate sector by examining their financial characteristics, funding behaviour, and lifecycle dynamics. Although zombie firms represent a relatively small share of firms, they account for a significant proportion of corporate debt and assets, indicating their broader implications for credit markets and economic efficiency. The findings show that zombification is generally associated with sustained deterioration in firm performance and increasing dependence on external financing. The study further highlights the strong associations between financing structure, governance characteristics, lender relationships, and zombie-firm persistence and recovery dynamics. Bank-dependent firms are associated with longer periods of financial distress, whereas equity-funded firms are associated with faster recovery and a lower likelihood of relapse. Overall, the results contribute to a better understanding of how financing patterns and institutional factors are associated with the survival, persistence, and recovery of zombie firms in the Indian corporate sector.

This study is subject to a few limitations common in the zombie firm literature. The identification of zombie firms relies on accounting-based indicators and threshold criteria, which may not fully capture underlying economic viability. Although alternative definitions and persistence conditions are considered, different measures may identify overlapping but not identical sets of firms leading to inconsistencies. The study is also subject to endogeneity concerns in interpreting the relationship between financing structure and zombification. Although lagged explanatory variables are employed to mitigate concerns related to reverse causality, they do not fully resolve the distinction between financing supply and firms' endogenous financing demand. Financially distressed firms may become increasingly dependent on bank financing after losing access to alternative sources of finance. Accordingly, the relationships reported in this study should be interpreted as conditional associations rather than evidence of causal effects. Moreover, while the observed differences across financing groups are consistent with several potential mechanisms, such as lender coordination, debt rollover, or governance-related effects, the present study does not directly identify or test these channels. Future studies should explore these relationships using stronger identification strategies to establish more robust causal evidence.

Nonetheless, the findings of this study carry important policy implications for financial stability, credit allocation, and productivity growth in India. The observed persistence of zombie firms, particularly among bank-dependent firms, is consistent with the financing patterns documented in this study. While these patterns may have implications for credit allocation and resource reallocation, the present analysis does not directly establish whether they arise from mechanisms such as continued lending, delayed restructuring, or other institutional factors. The results also indicate that fragmented banking relationships and weak governance structures are associated with more persistent financial distress. These associations may be relevant for credit monitoring and restructuring, although the underlying mechanisms are not directly examined in the present study. These findings highlight the need for stronger credit monitoring, better coordination among lenders, and more effective restructuring and insolvency mechanisms to ensure timely resolution of distressed firms. Overall, the study suggests that strengthening credit discipline and restructuring frameworks may improve the resolution of financially distressed firms and support more efficient capital allocation.⁵

⁵ Note: During the preparation of this work author(s) have used Chat-gpt to improve the language and readability. After using this tool /service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Appendix A: List of variables

NIC code	Three-digit National Industrial Classification Code
ICR	PBIT / Interest Expenses
Debt	Total Debt liability of the firm
Total Assets	Total assets owned by the firm
PBIT	Profit before Interest and Tax
NPM	Net Profit Margin
ROA	Return on Assets
ROE	Return on Equity
MC	Market Capitalisation of listed firms
Debt to Assets	Total Debt as a proportion of Total Assets
Leverage Median	Industry level median of Debt to Assets
NPM Median	Industry level median of Net Profit Margin
Debt Growth	Measure of change in Total Debt
Sales Growth	Measure of Change in Sales
Debt to Equity	Total Debt as a portion of firm's equity
Asset Disposal	Sale of Fixed Assets
Net Cashflow	Cash generated by the firm
Current Ratio	Ratio of current Assets to current liabilities
Borrowings from Banks	Long term Borrowings from Banks
Borrowings from Financial Institution	Long term Borrowings from Financial Institutions
Borrowings Syndicated across Banks and Financial Institutions	Long term Borrowings Syndicated across
Borrowings from Central and State Government	Long term Borrowings from Central & State Government
Inter-Corporate loan	Long term Borrowings between corporates
EDR	Earnings to debt Ratio
Loans from Promoters, Directors and Shareholders	Long term loans provided by the promoters, directors and the shareholders of the firm
Number of Banks	Number of banking relationships maintained by the firm

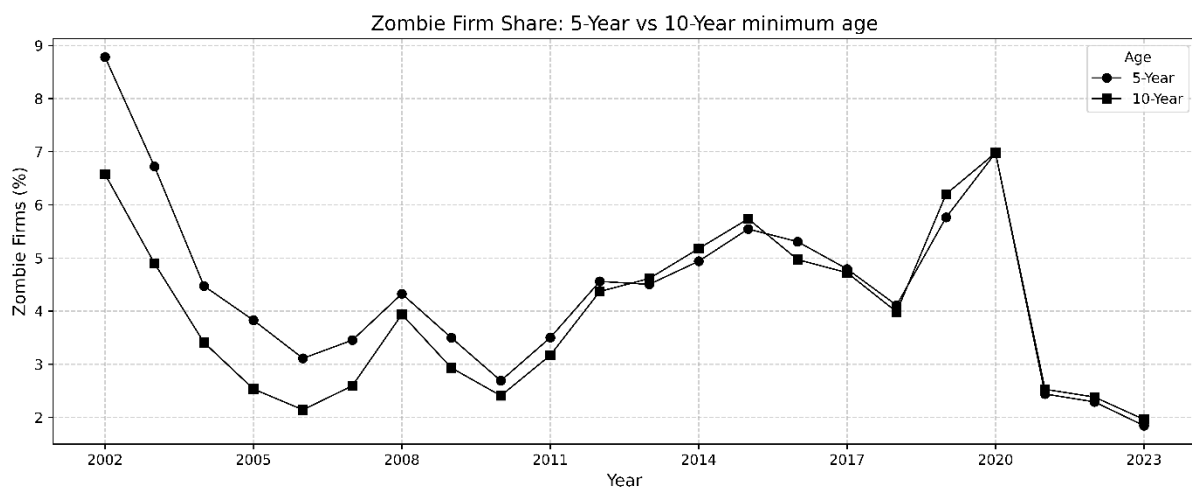
Interest Rate	Effective interest paid by the firm on its debt
Interest Rate Gap	Difference between the Interest rate paid by the firms and Prime lending rate of top five banks
Promoter Pledging	Percentage of shares pledged by promoter
Cashflow	Net cashflow of the firm
Altman Z-Score	Indicator of financial health and bankruptcy risk
Capex	Capital Expenditure of the firm
Tobins'q Ratio	Measure comparing a firm's market value to the replacement cost of its assets
Asset Disposal	Sale of fixed Asset
Primary Bank	Dummy for Firms Primarily funded by Banks
Primary Equity	Dummy for Firms Primarily funded by Equity

Appendix B: Robustness test results

Appendix B.1: Robustness check on minimum age for zombie firms

Zombie firms minimum age

Figure 20



Note: The semi-strong definition is used to compare the prevalence of zombie firms across samples restricted to firms with a minimum age of 5 years and 10 years.

Appendix B.2: Robustness check on definitions

Variables	Relaxed Definition			Restricted Definition		
	Model (1)	Model (2)	Model (3)	Model (1)	Model (2)	Model (3)
Primary Banks	0.713*** (0.028)	0.704*** (0.028)	0.499*** (0.033)	1.574*** (0.058)	1.552*** (0.058)	1.011*** (0.066)
Primary Equity	-1.252*** (0.078)	-1.30*** (0.078)	-1.278*** (0.086)	-1.306*** (0.206)	-1.453*** (0.206)	-1.188*** (0.213)
Number of Banks		0.047*** (0.005)	0.033*** (0.005)		0.096*** (0.009)	0.070*** (0.010)
Promoter pledging		0.010*** (0.001)	0.006*** (0.001)		0.016*** (0.002)	0.009*** (0.003)
Current ratio			-0.225*** (0.009)			-0.212*** (0.026)
Log Total Assets			0.107*** (0.013)			0.092 (0.053)
EDR			-1.654*** (0.026)			-.913*** (0.029)
Asset Disposal			0.051*** (0.013)			-0.081 (0.038)
ROA			-0.038*** (0.001)			-0.073*** (0.002)
Cashflow			-0.010*** (0.002)			-0.010* (0.005)
Age			-0.299*** (0.018)			0.458*** (0.041)
Constant	-2.812*** (0.054)	-2.803*** (0.067)	-1.718*** (0.091)	-7.21*** (0.067)	-7.27** (0.149)	-3.370*** (0.110)
Estimator	MLE	MLE	MLE	MLE	MLE	MLE
Firm controls	No	No	Yes	No	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	264,988	264,988	207,793	260,330	260,330	204,404
Number of firms	23,637	23,637	23,146	23,637	23,637	23,094

Appendix C: Log Rank Test Results

Appendix C.1: Log rank test results for Kaplan-Meier Estimates by Primary Source

	Test Statistic	P-Value
Zombie Continuation	14.57	0.000
Non-Zombie Persistence	1138.66	0.000
Zombie Recovery	12.87	0.000
Zombie Relapse	230.01	0.000

Appendix C.2 Log Rank test results for Business Group Affiliation

	Test Statistic	P-Value
Zombie Continuation	15.9	0.000
Non-Zombie Persistence	20.61	0.000
Zombie Recovery	14.45	0.000
Zombie Relapse	4.86	0.030

Appendix C.3 Log Rank test results for Business Group Affiliation for Listed Firms

	Test Statistic	P-Value
Zombie Continuation	0.63	0.43
Non-Zombie Persistence	105.52	0.000
Zombie Recovery	0.30	0.59
Zombie Relapse	1.66	0.20

Appendix C.4 Log Rank Test results for Kaplan-Meier Estimates by Sector

	Test Statistic	P-Value
Zombie Continuation	48.03	0.000
Non-Zombie Persistence	20.61	0.000
Zombie Recovery	26.91	0.000
Zombie Relapse	4.86	0.77

Appendix D: Transition Matrices

Appendix D.1: Zombie Entry Matrix

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
2002		476	160	72	35	27	19	14	12	9	5	4	4	2	2	1	1	1	1	1	1	0
2003			296	64	28	13	8	8	3	3	3	3	1	1	1	1	0	0	0	0	0	0
2004				208	48	23	15	11	11	6	6	3	3	3	3	3	2	2	2	1	0	0
2005					237	47	13	4	3	2	1	1	0	0	0	0	0	0	0	0	0	0
2006						204	57	19	10	6	4	3	2	1	1	1	1	0	0	0	0	0
2007							247	51	15	8	5	5	1	1	1	1	1	0	0	0	0	0
2008								346	62	19	12	11	8	6	4	3	1	1	1	0	0	0
2009									279	50	14	5	5	3	2	2	1	0	0	0	0	0
2010										228	54	27	11	6	6	3	2	2	2	1	0	0
2011											315	77	31	18	9	7	5	4	3	2	1	0
2012												393	115	41	22	13	8	6	4	1	0	0
2013													433	113	50	26	17	13	9	2	1	0
2014														594	192	75	32	15	7	6	4	1
2015															706	199	70	37	18	10	4	0
2016																675	157	42	24	11	7	2
2017																	636	149	52	27	7	4
2018																		574	165	56	11	3
2019																			892	260	41	13
2020																				1013	121	14
2021																					305	59
2022																						333
2023																						

Appendix D.2: Zombie Recovery Matrix

	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
2002	0	290	80	32	7	8	5	2	3	4	1	0	2	0	1	0	0	0	0	0	0	1
2003		0	217	31	14	5	0	5	0	0	0	0	2	0	0	0	1	0	0	0	0	0
2004			0	153	21	7	3	0	4	0	0	3	0	0	0	0	1	0	0	1	1	0
2005				0	179	32	7	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0
2006					0	134	31	9	4	2	1	1	1	0	0	0	0	1	0	0	0	0
2007						0	181	34	7	3	0	0	4	0	0	0	0	0	0	0	0	0
2008							0	271	40	6	1	3	2	2	1	1	0	0	0	1	0	0
2009								0	220	35	8	0	2	1	0	1	1	0	0	0	0	0
2010									0	169	26	16	4	0	3	1	0	0	1	1	0	0
2011										0	226	43	13	8	2	2	1	1	1	1	0	1
2012											0	274	70	19	8	5	2	2	3	1	0	0
2013												0	305	62	24	7	4	2	7	1	1	0
2014													0	389	112	40	17	6	1	2	3	0
2015														0	482	119	26	19	8	6	4	0
2016															0	485	102	18	13	3	5	1
2017																0	461	85	22	20	3	3
2018																	0	370	101	42	8	2
2019																		0	585	209	26	9
2020																			0	849	103	12
2021																				0	230	48
2022																					0	295
2023																						0

Appendix D.3: Zombie Exit Matrix

	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
2002		0	26	8	5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2003			0	15	5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2004				0	7	4	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
2005					0	11	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2006						0	13	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2007							0	15	2	0	0	0	0	0	0	0	0	1	0	0	0	0
2008								0	13	3	1	0	0	0	0	1	0	0	0	0	0	0
2009									0	9	1	1	0	0	0	0	0	0	0	0	0	0
2010										0	5	1	0	1	0	0	0	0	0	0	0	0
2011											0	12	3	0	1	0	0	0	0	0	0	0
2012												0	4	4	0	1	0	0	0	0	0	0
2013													0	15	1	0	2	0	2	0	0	0
2014														0	13	5	3	0	2	0	0	0
2015															0	25	10	7	0	0	0	0
2016																0	33	13	0	0	1	0
2017																	0	26	12	3	0	0
2018																		0	39	8	3	0
2019																			0	47	10	2
2020																				0	43	4
2021																					0	16
2022																						0
2023																						

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